

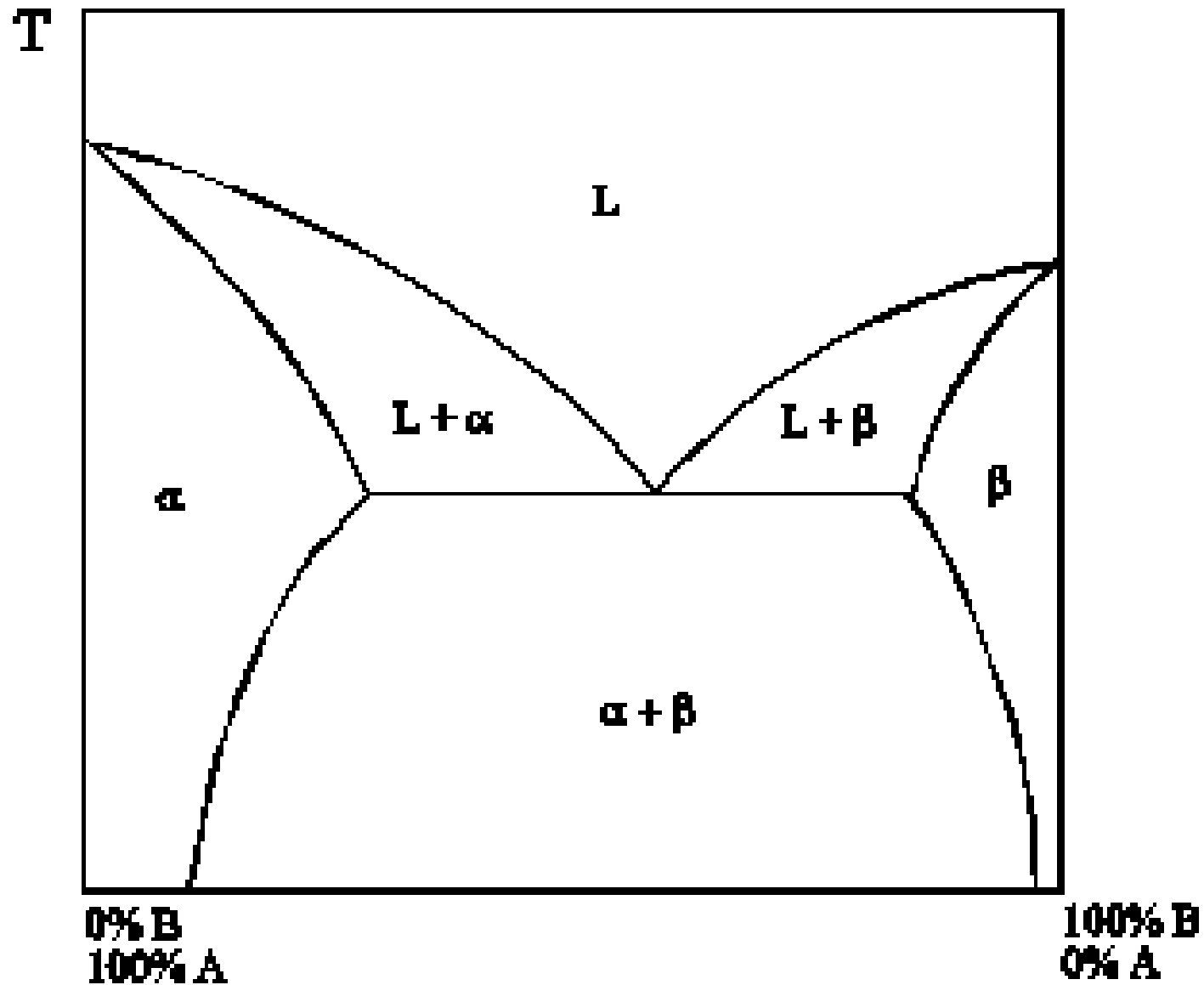


STM Exercise 7

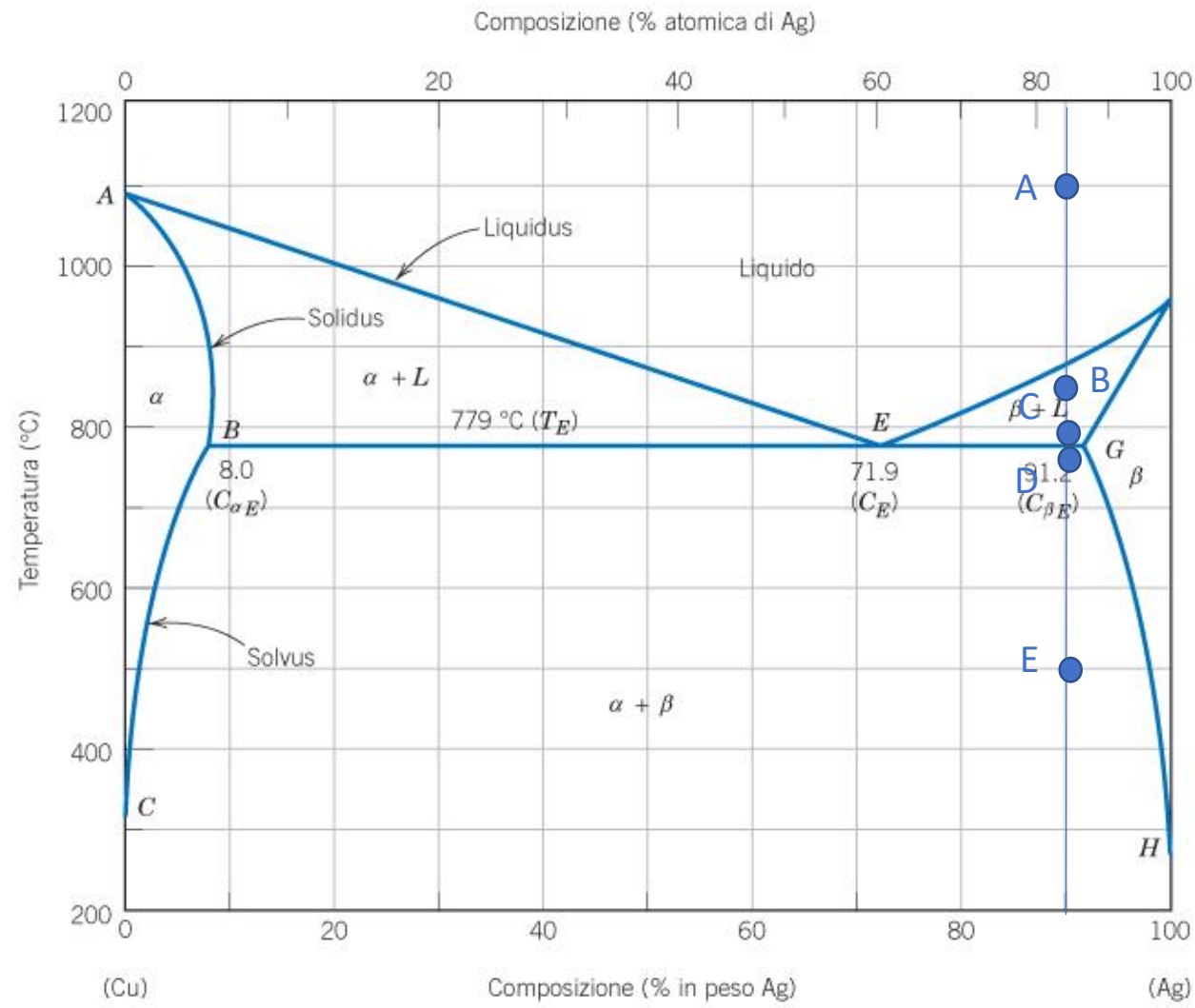
Phase Diagrams
(partial solubility in the solid state)

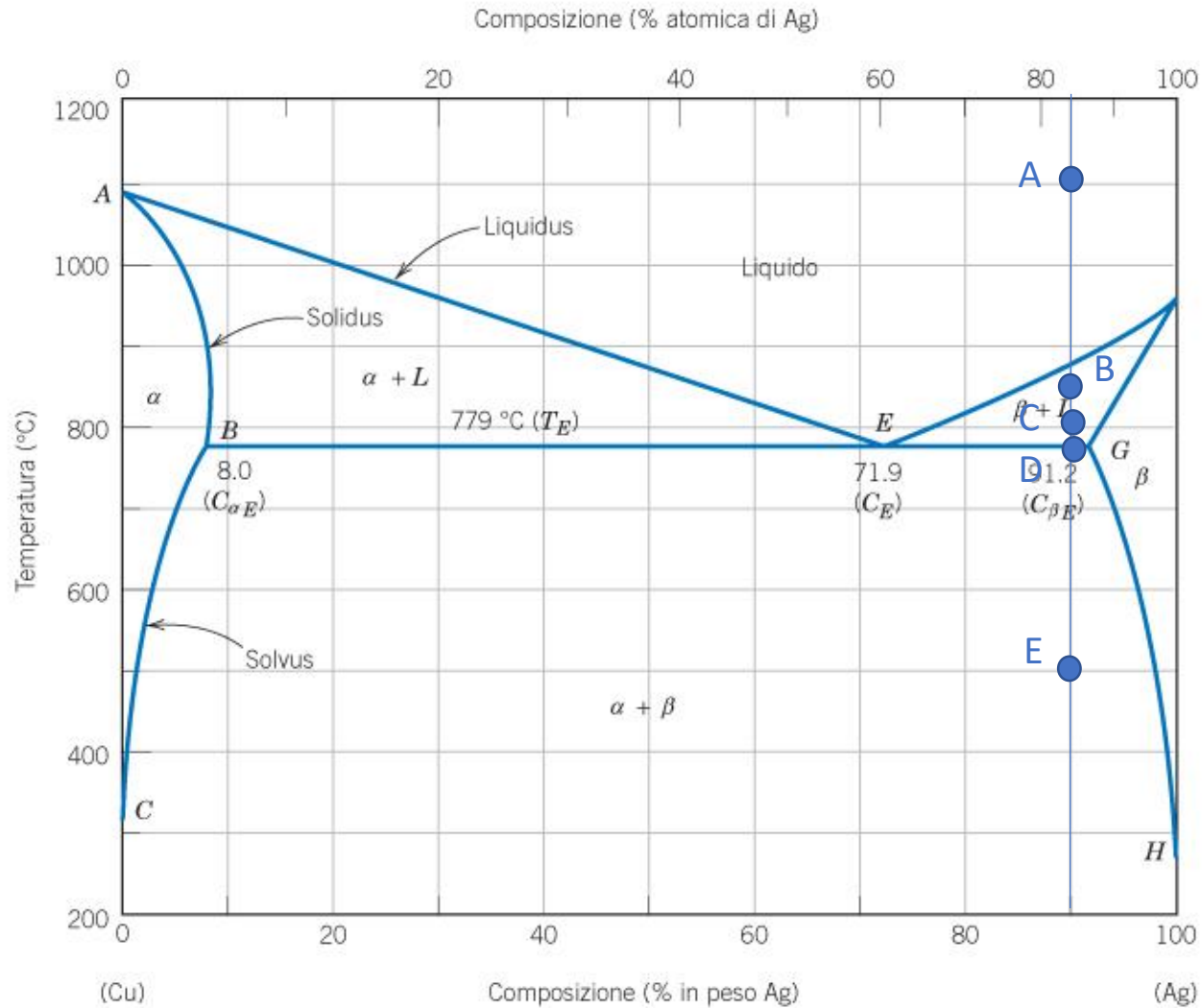
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Complete solubility in the liquid state,
partial solubility in the solid state



Exercise 1: determine the phase present, their composition and relative amounts in the points **A**, **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



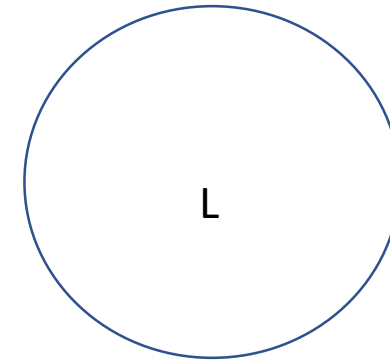


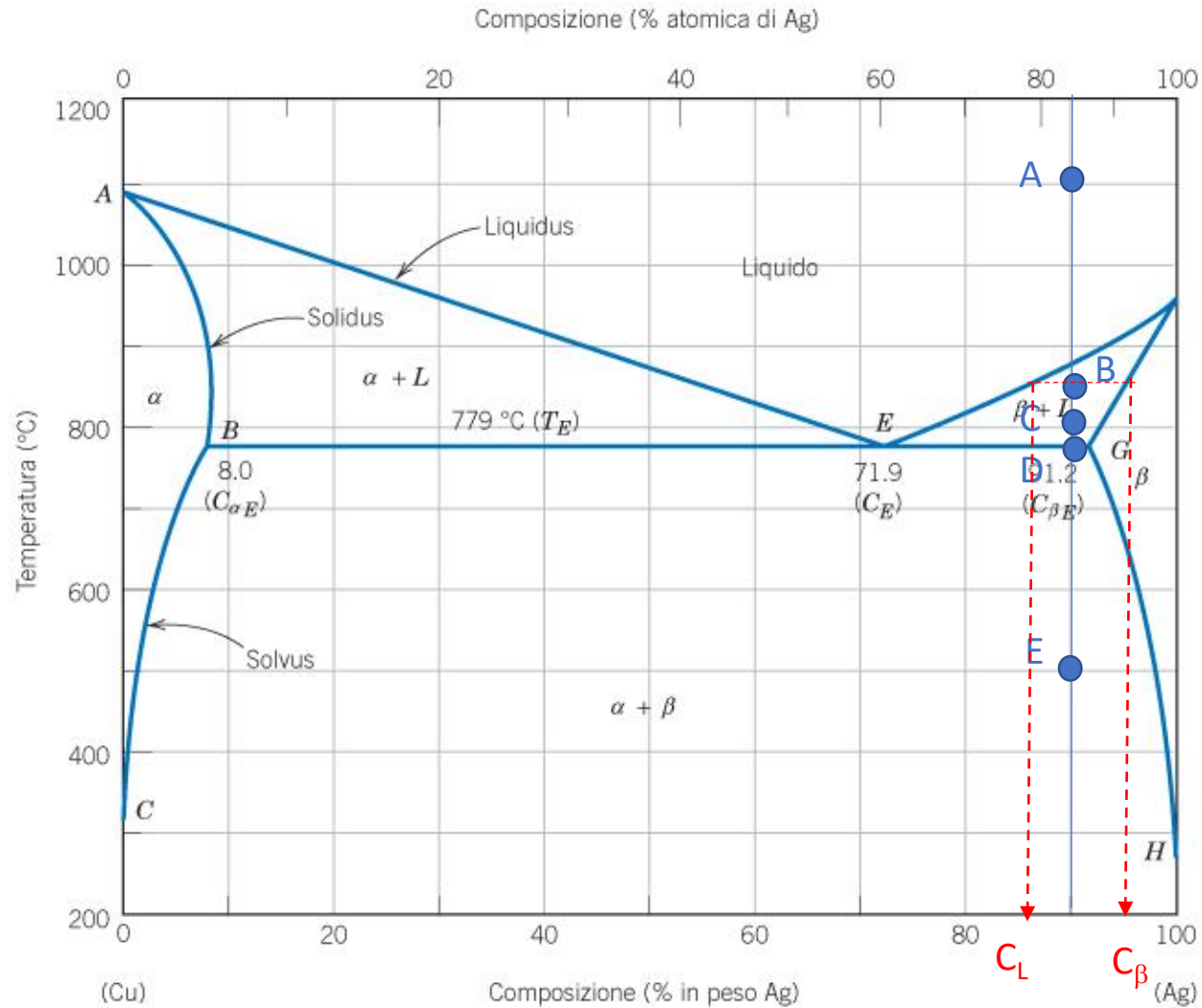
Point A:

Phases: 1 phase liquid

Composition: 90%Ag, 10%Cu

Relative amount: 100% liquid phase





Point B:

Phases: 2 phases solid (β) + liquid (L)

Composition: 2 phases region → horizontal rule

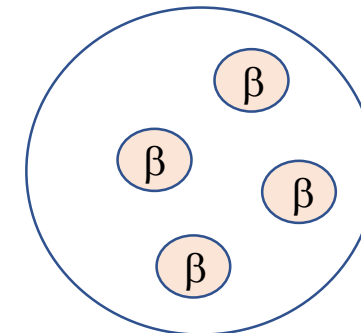
Composition of liquid (L): $C_L = 86\% \text{ Ag}$, 14% Cu

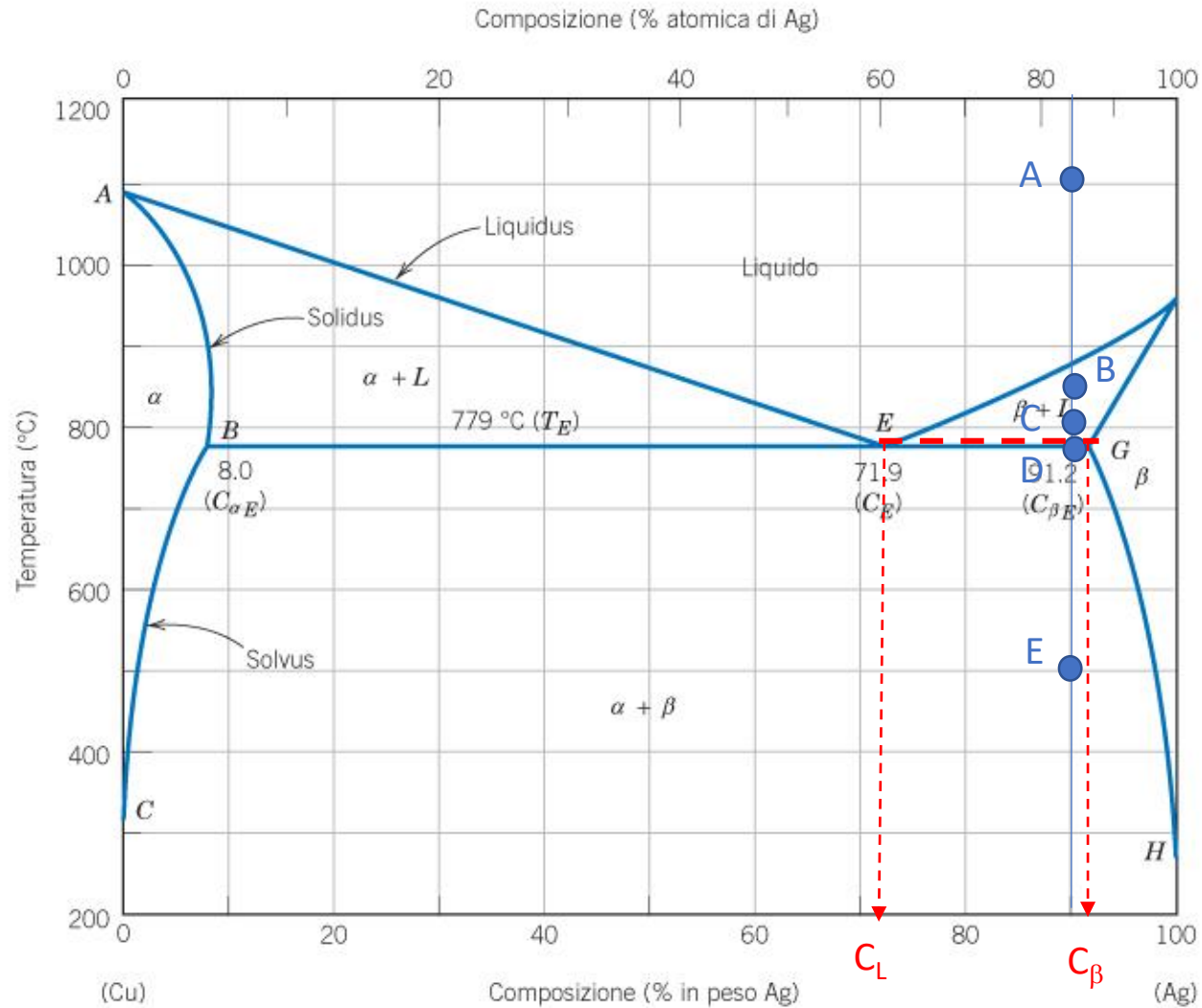
Composition of solid (β): $C_{\beta} = 95\% \text{ Ag}$, 5% Cu

Relative amount: 2 phases region → lever rule

$$\% \beta = (90 - 86) / (95 - 86) * 100 = 44\%$$

$$\% L = (95 - 90) / (95 - 86) * 100 = 56\%$$





Point C:

Phases: 2 phases solid (β) + liquid (L)

Composition: 2 phases region → horizontal rule

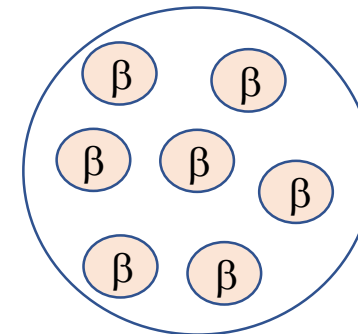
Composition of liquid (L): $C_L = 71.9\% \text{ Ag}$, $28.1\% \text{ Cu}$

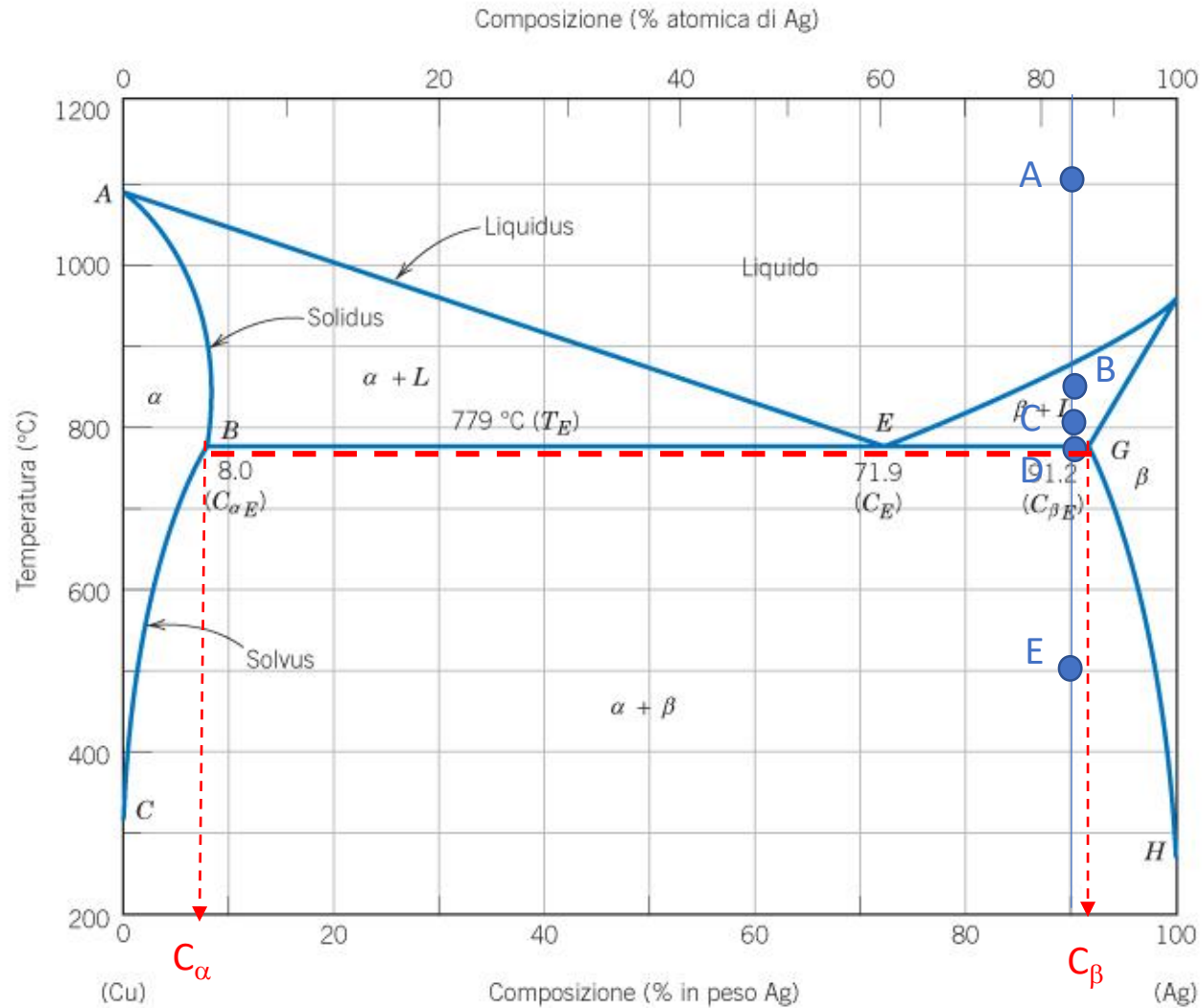
Composition of solid (β): $C_{\beta} = 91.2\% \text{ Ag}$, $8.8\% \text{ Cu}$

Relative amount: 2 phases region → lever rule

$$\% \beta = (90 - 71.9) / (91.2 - 71.9) * 100 = 93.8\%$$

$$\% L = (91.2 - 90) / (91.2 - 71.9) * 100 = 6.2\%$$





Point D:

Phases: 2 phases solid (α) + solid (β)

Composition: 2 phases region → horizontal rule

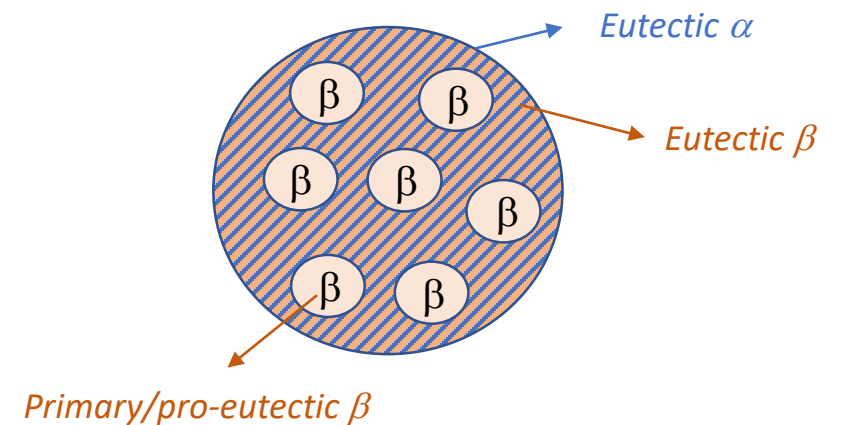
Composition of solid (α): $C_{\alpha} = 8\% \text{Ag}, 92\% \text{Cu}$

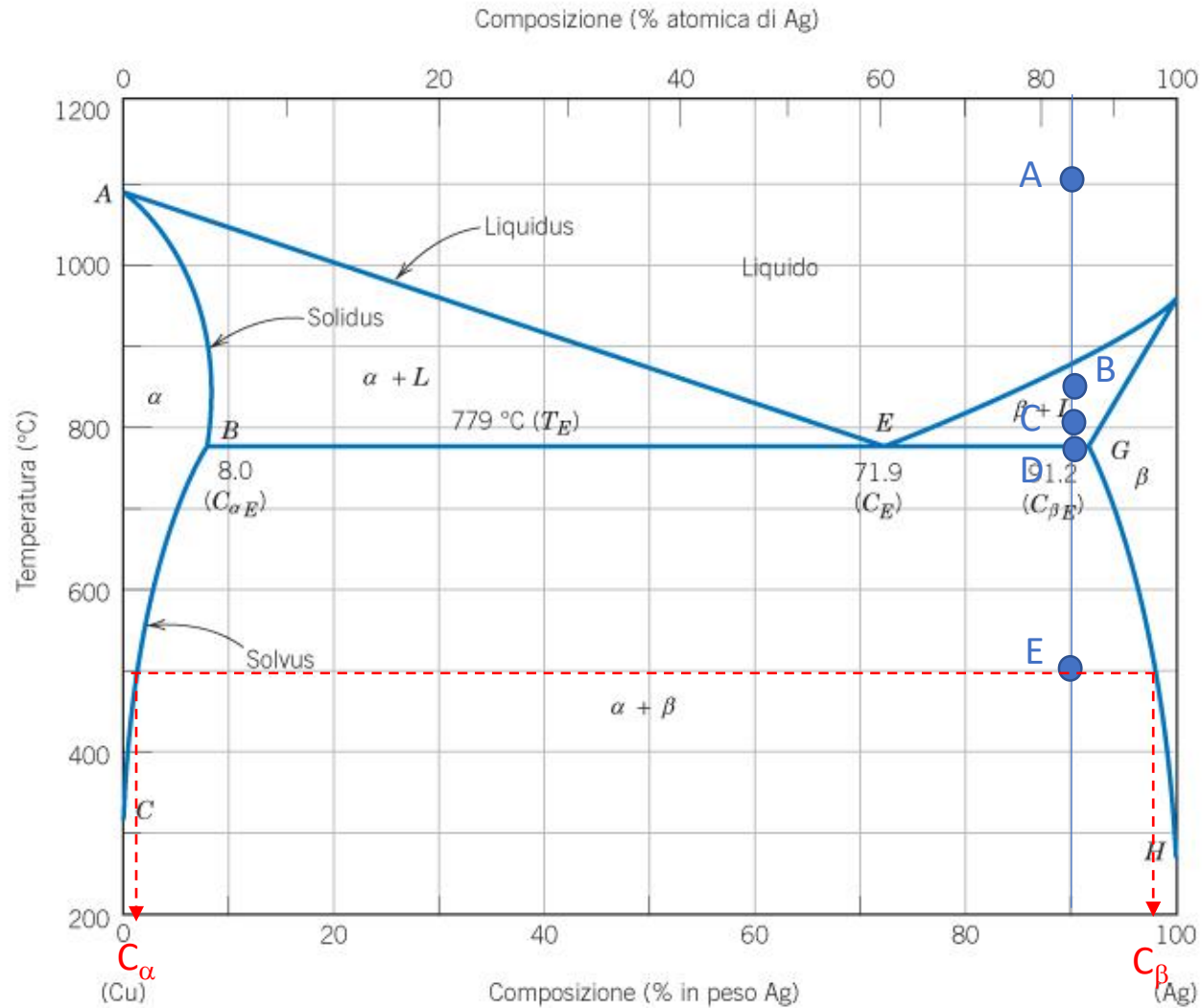
Composition of solid (β): $C_{\beta} = 91.2\% \text{Ag}, 8.8\% \text{Cu}$

Relative amount: 2 phases region → lever rule

$$\% \alpha = (91.2 - 90) / (91.2 - 8) * 100 = 1.4\%$$

$$\% \beta = (90 - 8) / (91.2 - 8) * 100 = 98.6\%$$





Point E:

Phases: 2 phases solid (α) + solid (β)

Composition: 2 phases region → horizontal rule

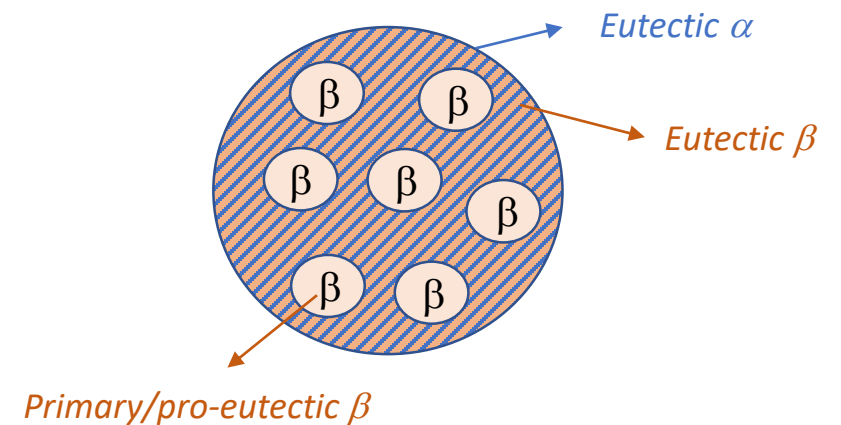
Composition of solid (α): $C_{\alpha} = 2\% \text{Ag}, 98\% \text{Cu}$

Composition of solid (β): $C_{\beta} = 98\% \text{Ag}, 2\% \text{Cu}$

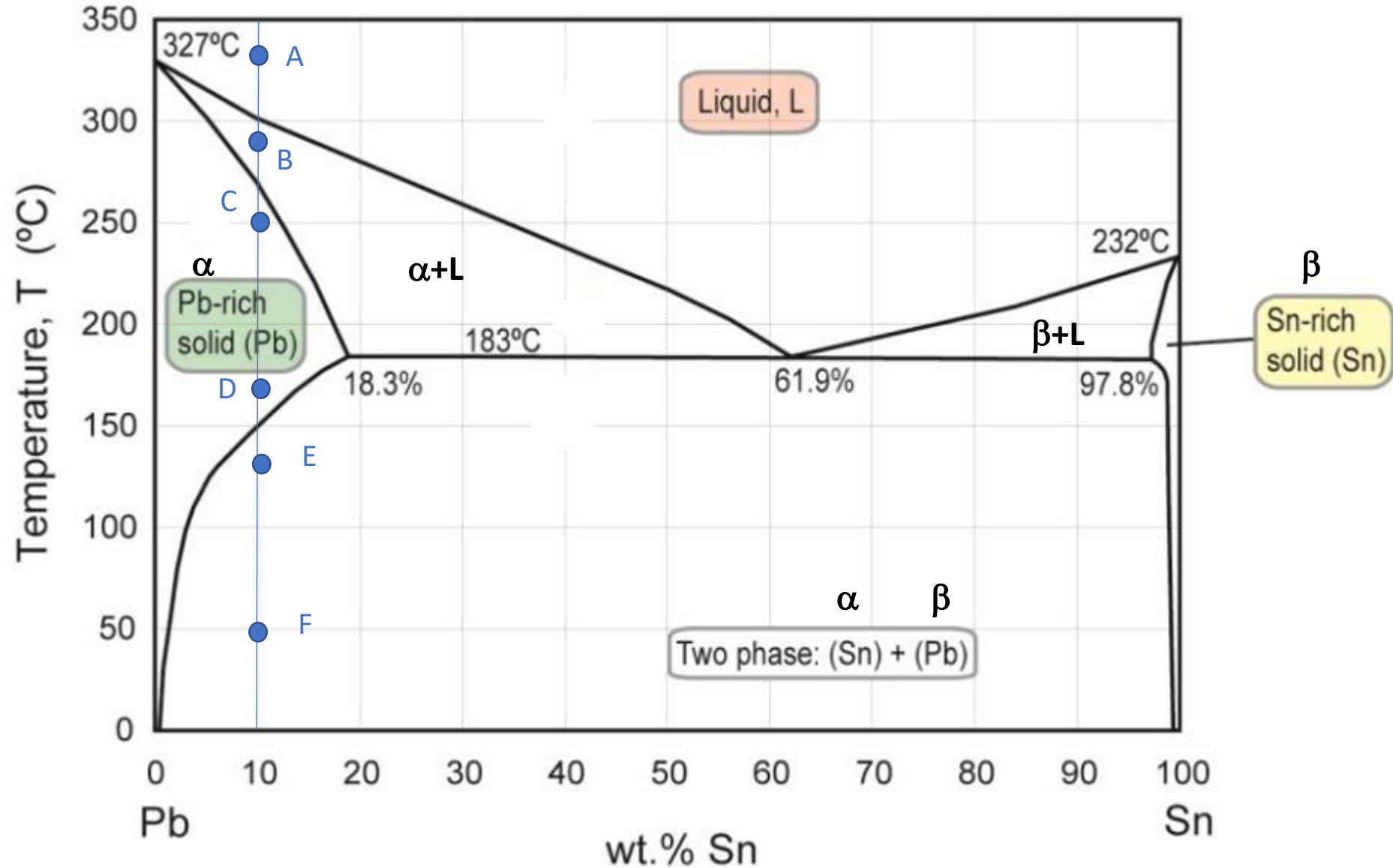
Relative amount: 2 phases region → lever rule

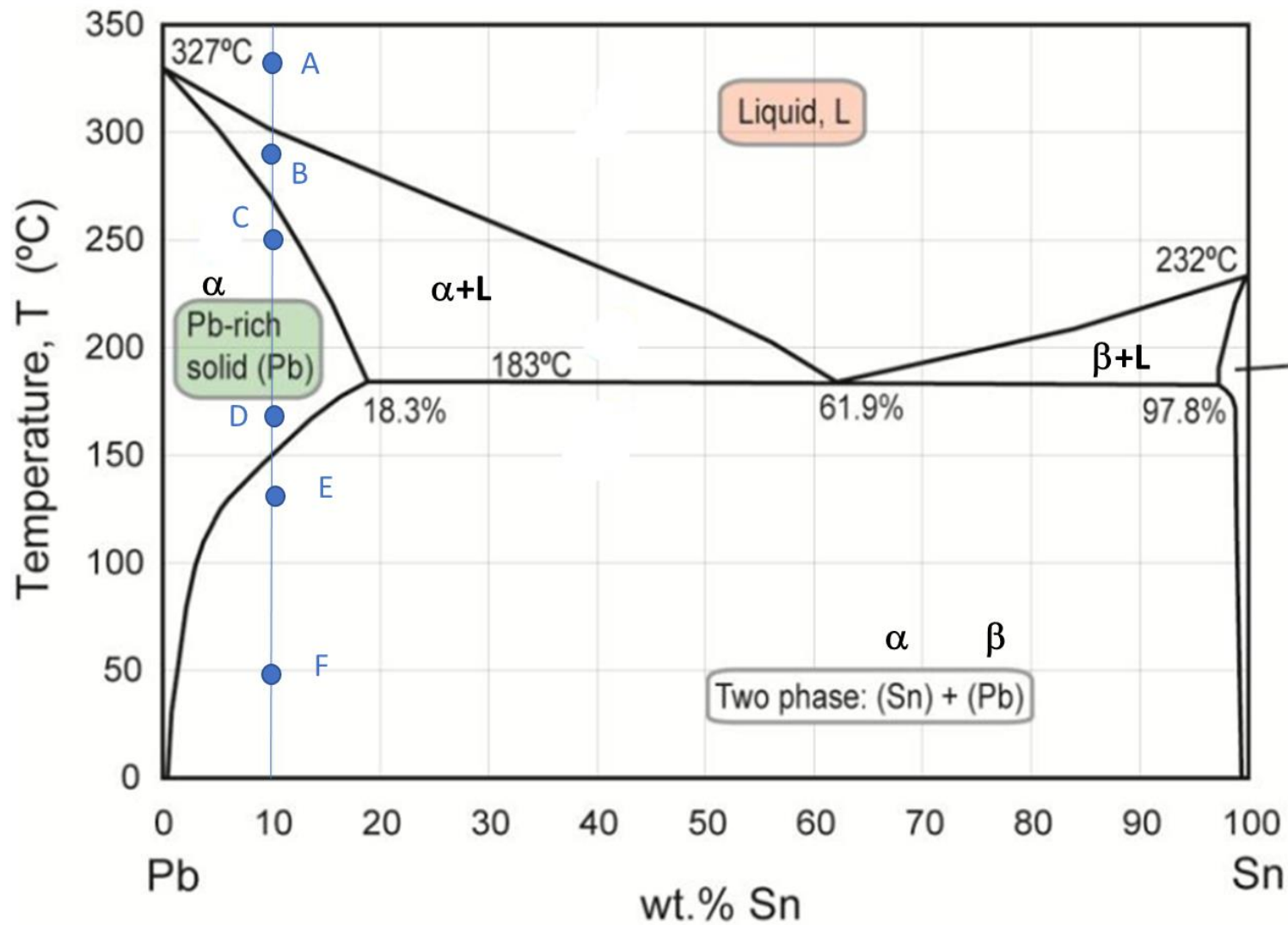
$$\% \alpha = (98 - 90) / (98 - 2) * 100 = 8.3\%$$

$$\% \beta = (90 - 2) / (98 - 2) * 100 = 91.7\%$$



Exercise 2: determine the phase present, their composition and relative amounts in the points A, B, C, D, E and F indicated in the phase diagram. Schematize the microstructure in each point.



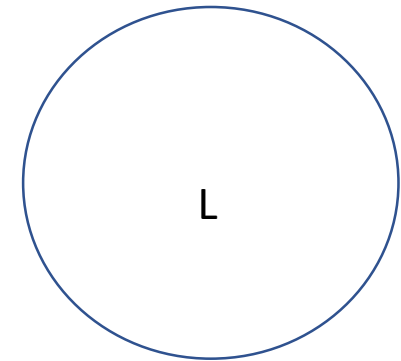


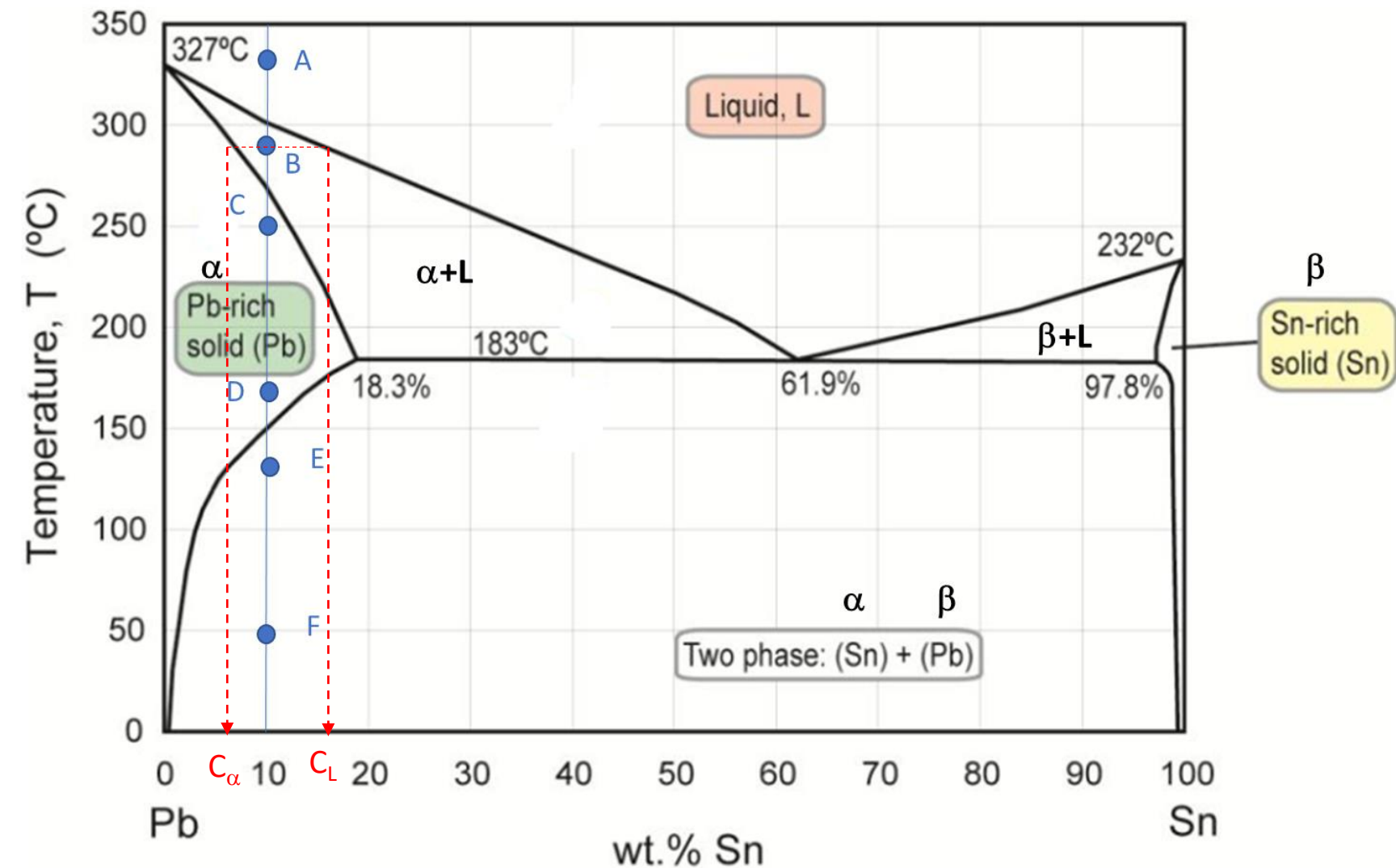
Point A:

Phases: 1 phase liquid

Composition: 10% Sn, 90% Pb

Relative amount: 100% liquid phase





Point B:

Phases: 2 phases solid (α) + liquid (L)

Composition: 2 phases region $\rightarrow$ horizontal rule

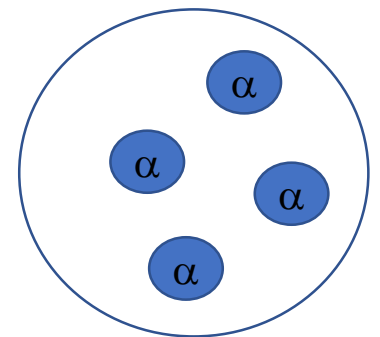
Composition of liquid (L): $C_L = 17\% \text{Sn}, 83\% \text{Pb}$

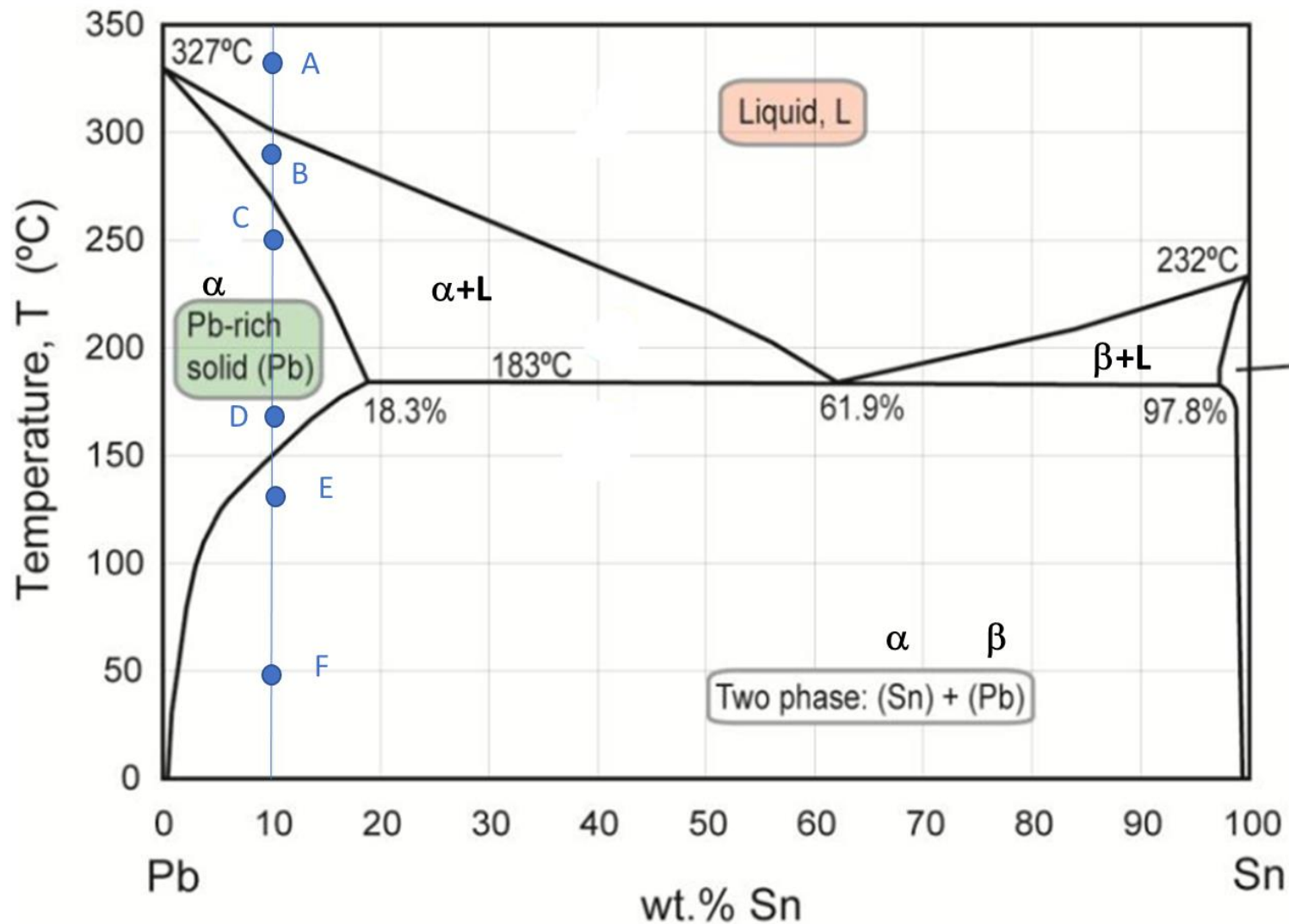
Composition of solid (α): $C_{\alpha} = 7\% \text{Sn}, 93\% \text{Pb}$

Relative amount: 2 phases region $\rightarrow$ lever rule

$\% \alpha = (17 - 10) / (17 - 7) * 100 = 70\%$

$\% L = (10 - 7) / (17 - 7) * 100 = 30\%$





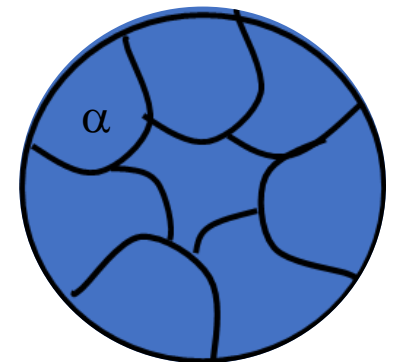
Point C:

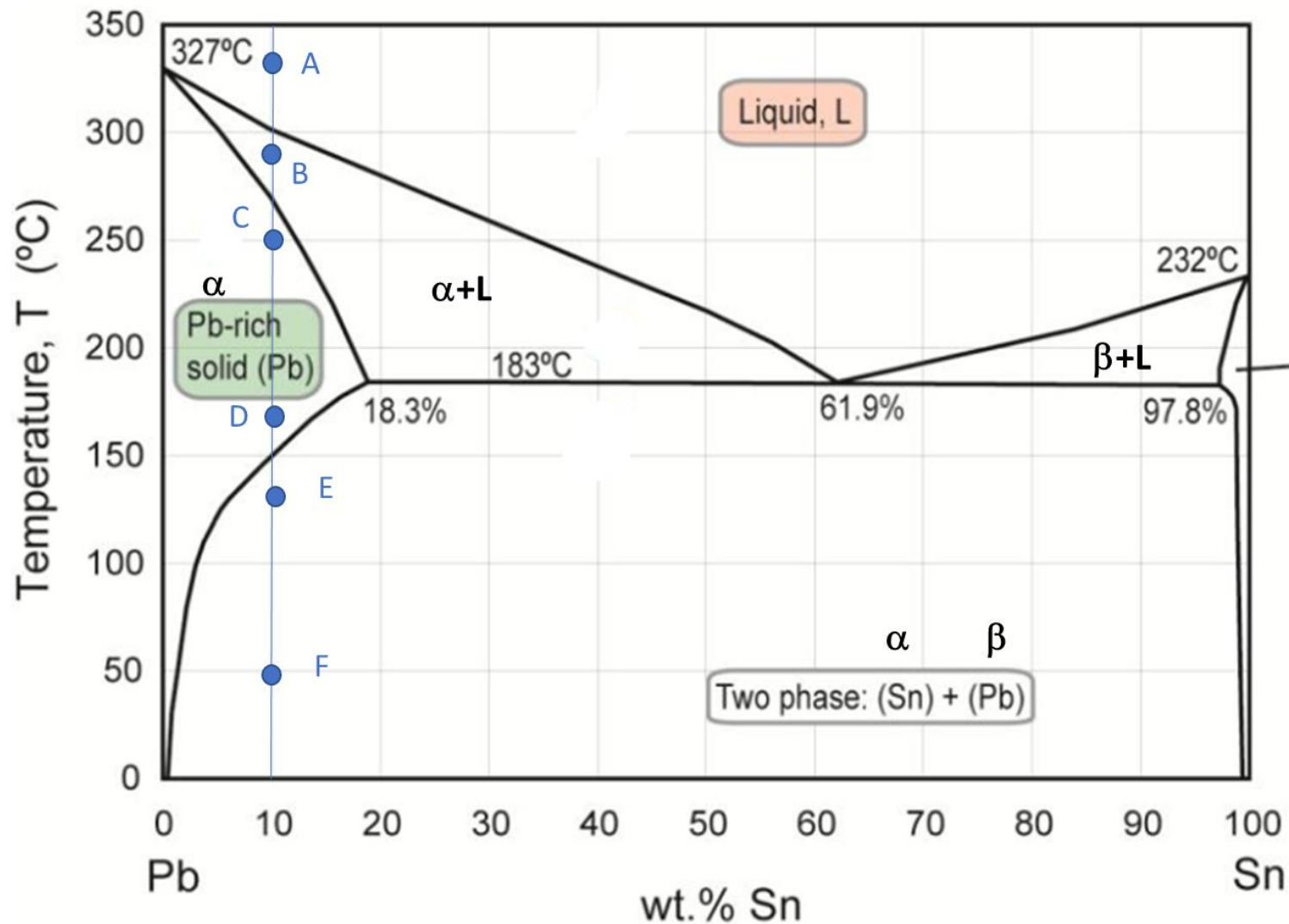
Phases: 1 phase solid (α)

Composition: 10% Sn, 90% Pb

Relative amount: 100% solid phase α

β
Sn-rich
solid (Sn)





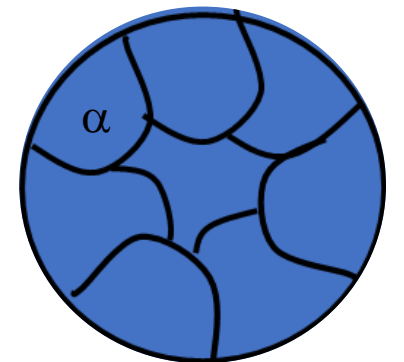
Point D:

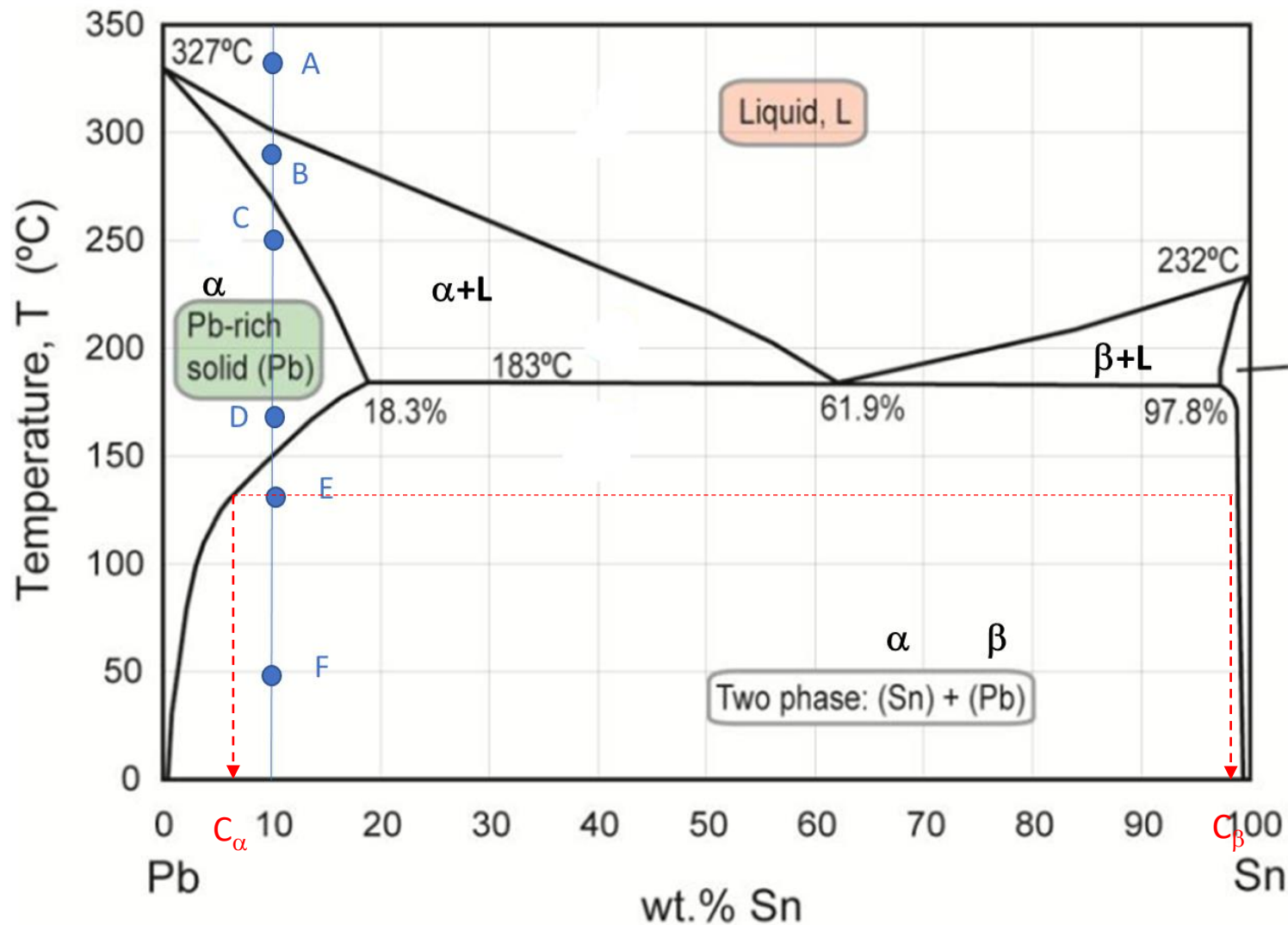
Phases: 1 phase solid (α)

Composition: 10% Sn, 90% Pb

Relative amount: 100% solid phase α

β
Sn-rich
solid (Sn)





Point E:

Phases: 2 phases solid (α) + solid (β)

Composition: 2 phases region → horizontal rule

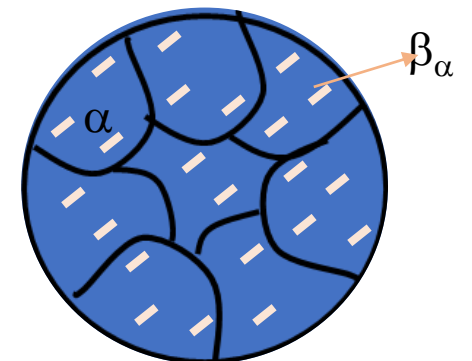
Composition of solid (α): $C_\alpha = 7\%$ Sn, 93% Pb

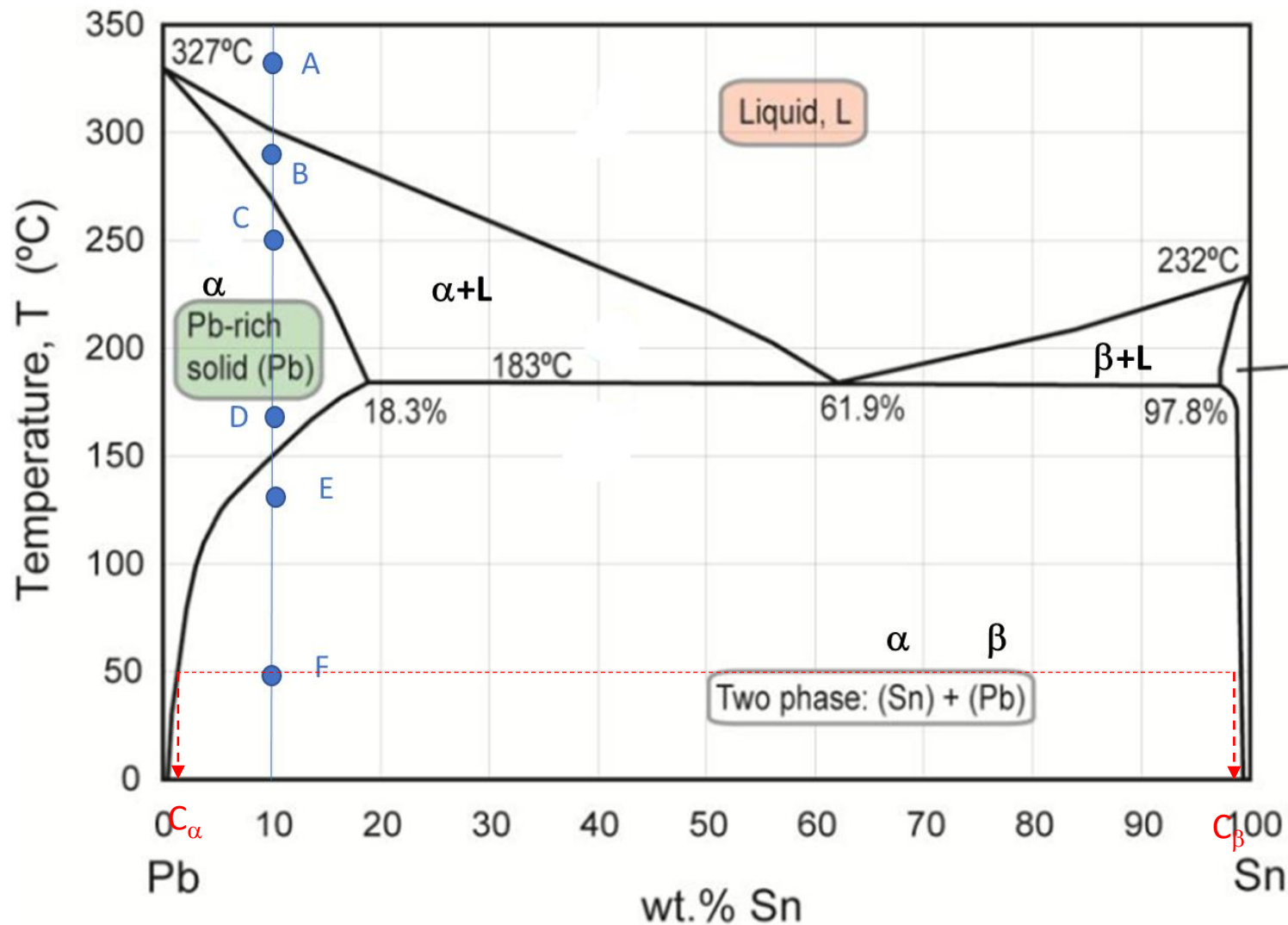
Composition of solid (β): $C_\beta = 98\%$ Sn, 2% Pb

Relative amount: 2 phases region → lever rule

$$\% \alpha = (98 - 10) / (98 - 7) * 100 = 96.7\%$$

$$\% \beta = (10 - 7) / (98 - 7) * 100 = 3.3\%$$





Point F:

Phases: 2 phases solid (α) + solid (β)

Composition: 2 phases region → horizontal rule

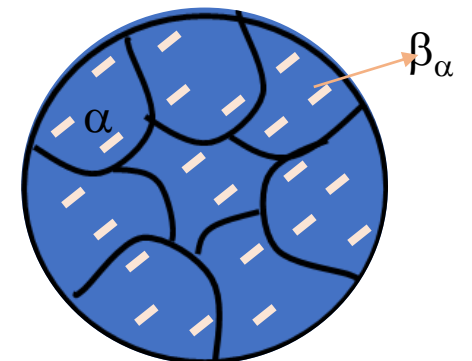
Composition of solid (α): $C_\alpha = 2\% \text{Sn}, 98\% \text{Pb}$

Composition of solid (β): $C_\beta = 99\% \text{Sn}, 1\% \text{Pb}$

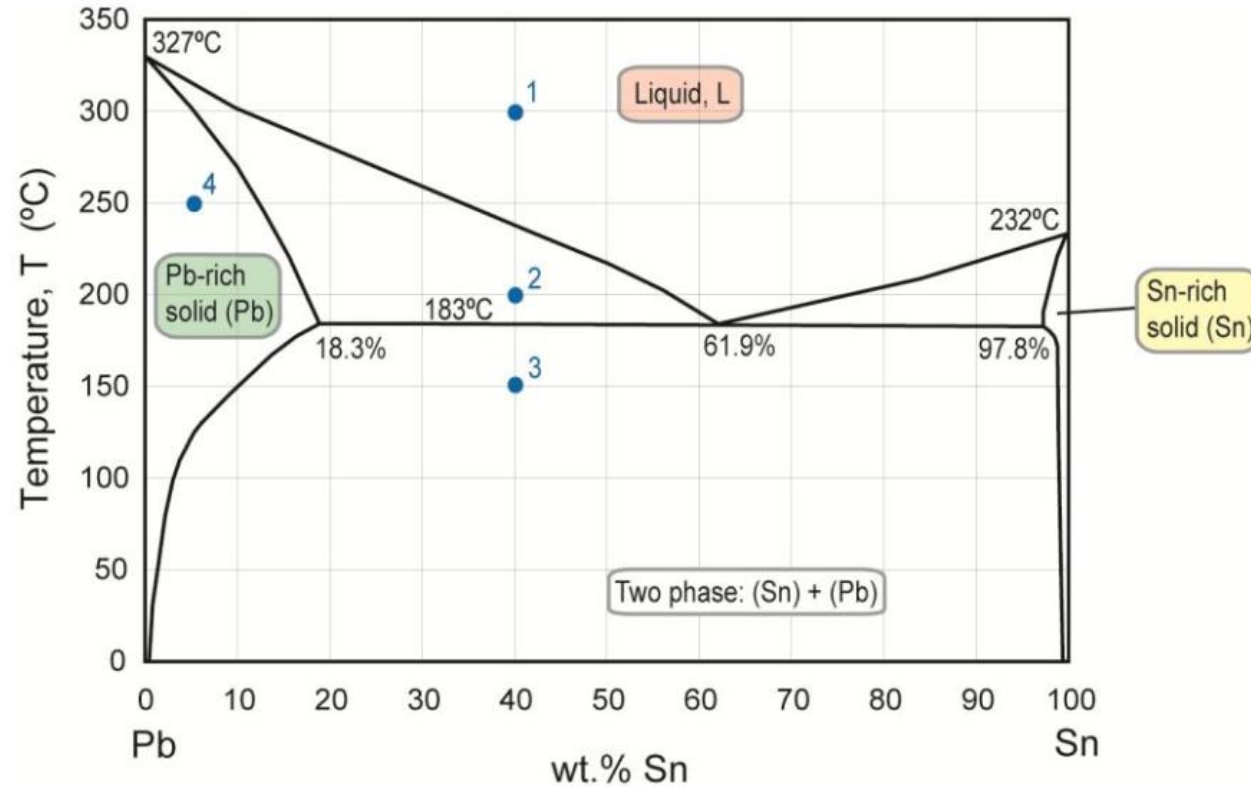
Relative amount: 2 phases region → lever rule

$$\% \alpha = (99 - 10) / (99 - 2) * 100 = 91.7\%$$

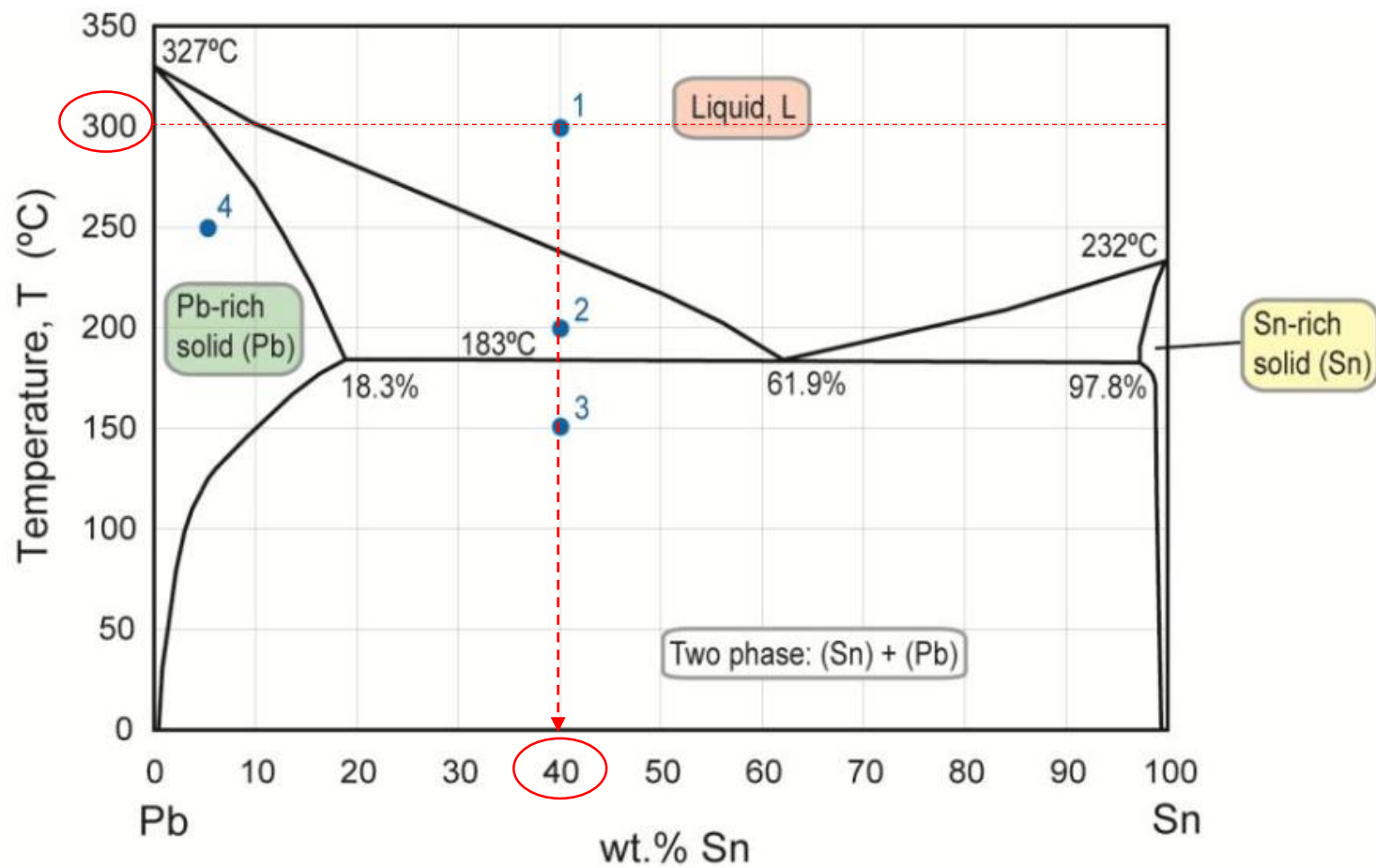
$$\% \beta = (10 - 2) / (99 - 2) * 100 = 8.3\%$$



E.6 Use the Pb-Sn diagram below to answer the following questions.



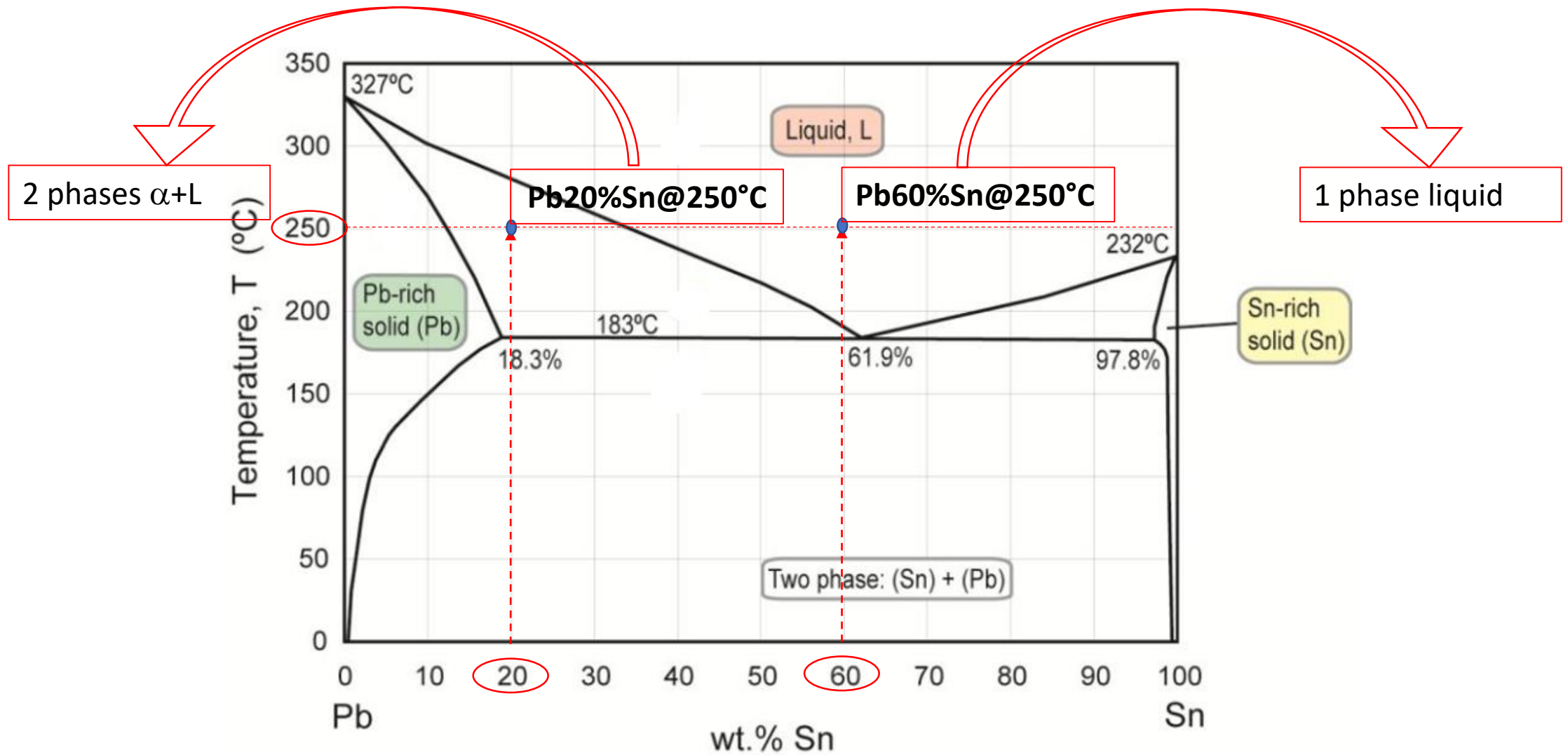
- (a) What are the values of the state variables (composition and temperature) at constitution point 1?
- (b) Mark the constitution points for Pb-60wt% Sn and Pb-20wt% Sn alloys at 250°C. What phases are present in each case?
- (c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?
- (d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:
 - (i) the initial composition, temperature and phase(s);
 - (ii) the temperature at which a phase change occurs, and the final phase(s).



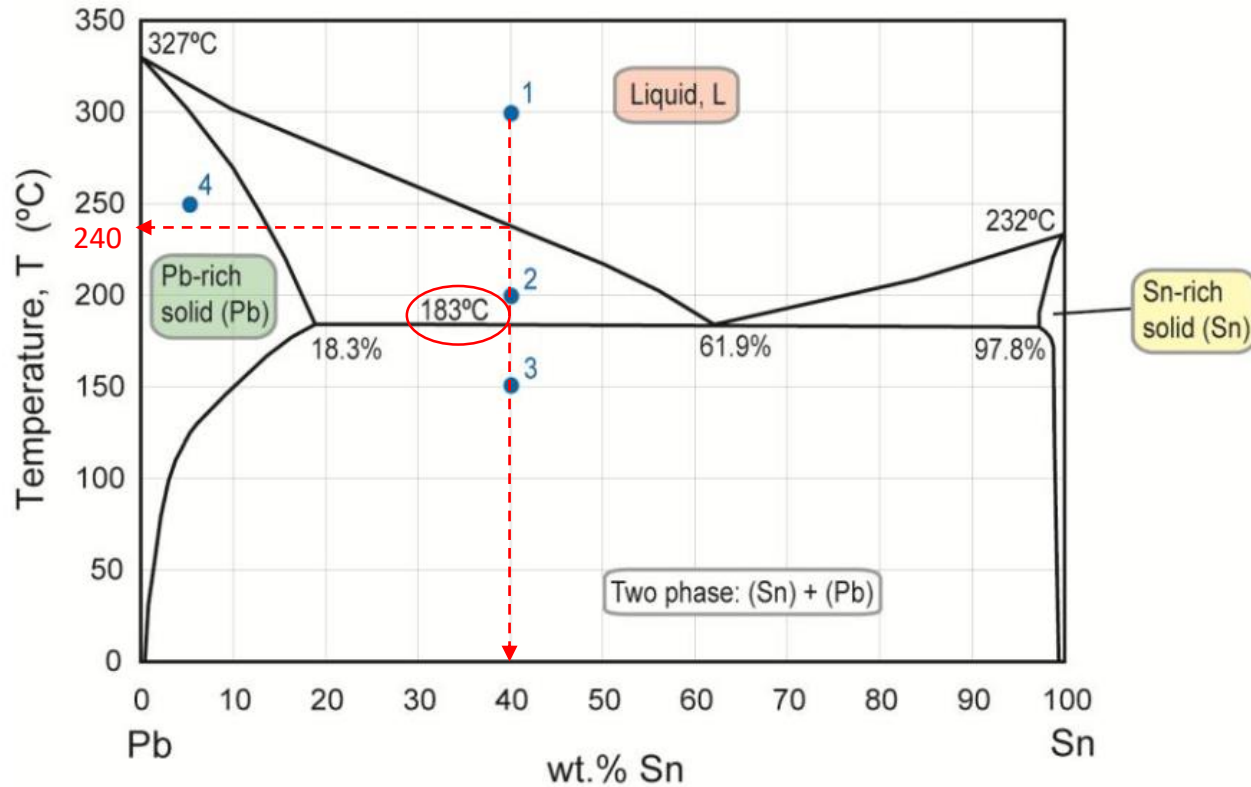
Point 1

1 phase liquid, composition 40%Sn,
60% Pb

Temperature 300°C



(c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?



Alloy 1:

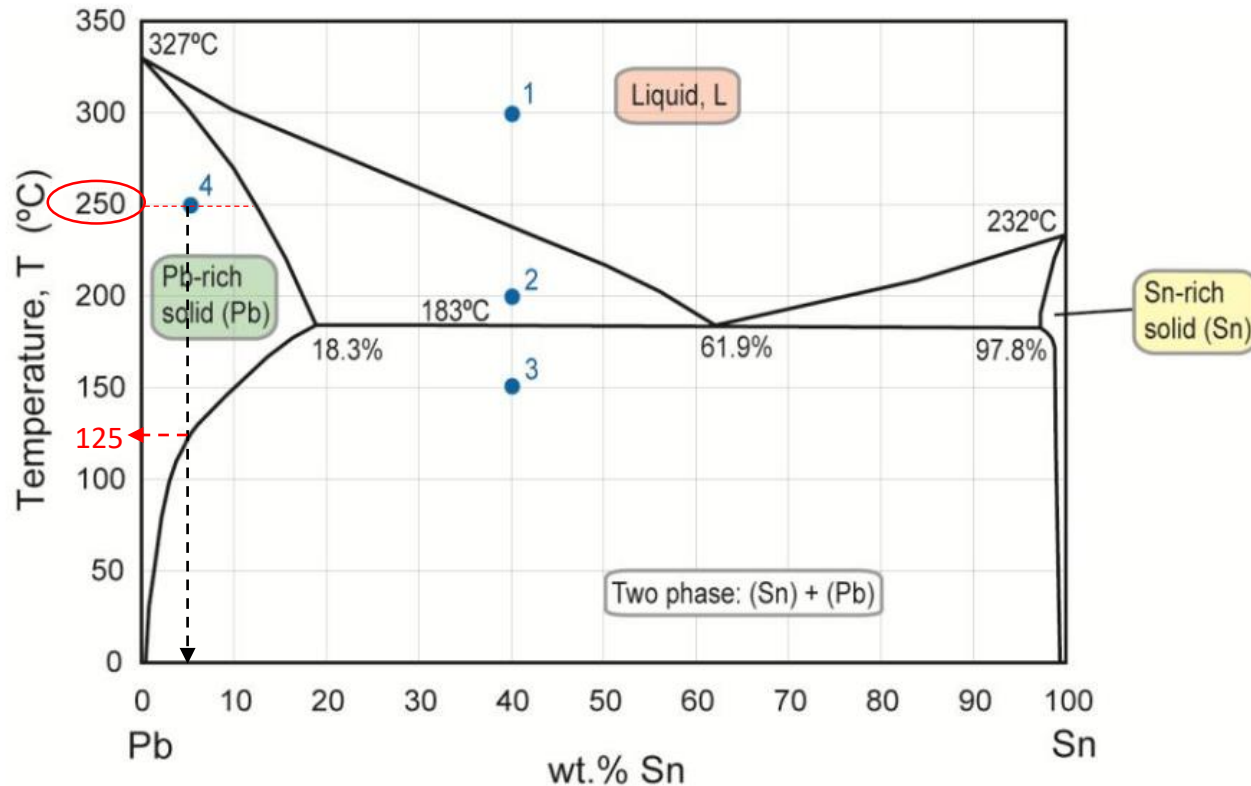
- 300°C – only liquid
- 240°C - solidification of the first α crystal
- 183°C - all the liquid undergoes eutectic transformation producing α - β lamellar microstructure

Point 2: 2 phases α (solid rich in Pb) and Liquid (L)

Point 3: 2 phases α (solid rich in Pb) and β (solid rich in Sn)

(d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:

- (i) the initial composition, temperature and phase(s);
- (ii) the temperature at which a phase change occurs, and the final phase(s).

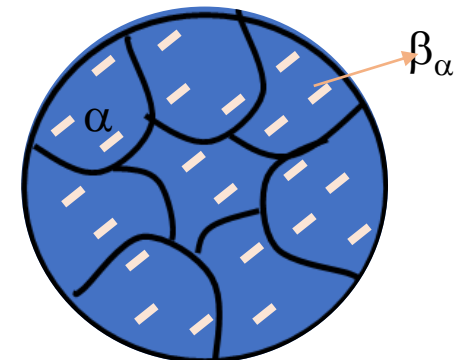


Alloy 4:

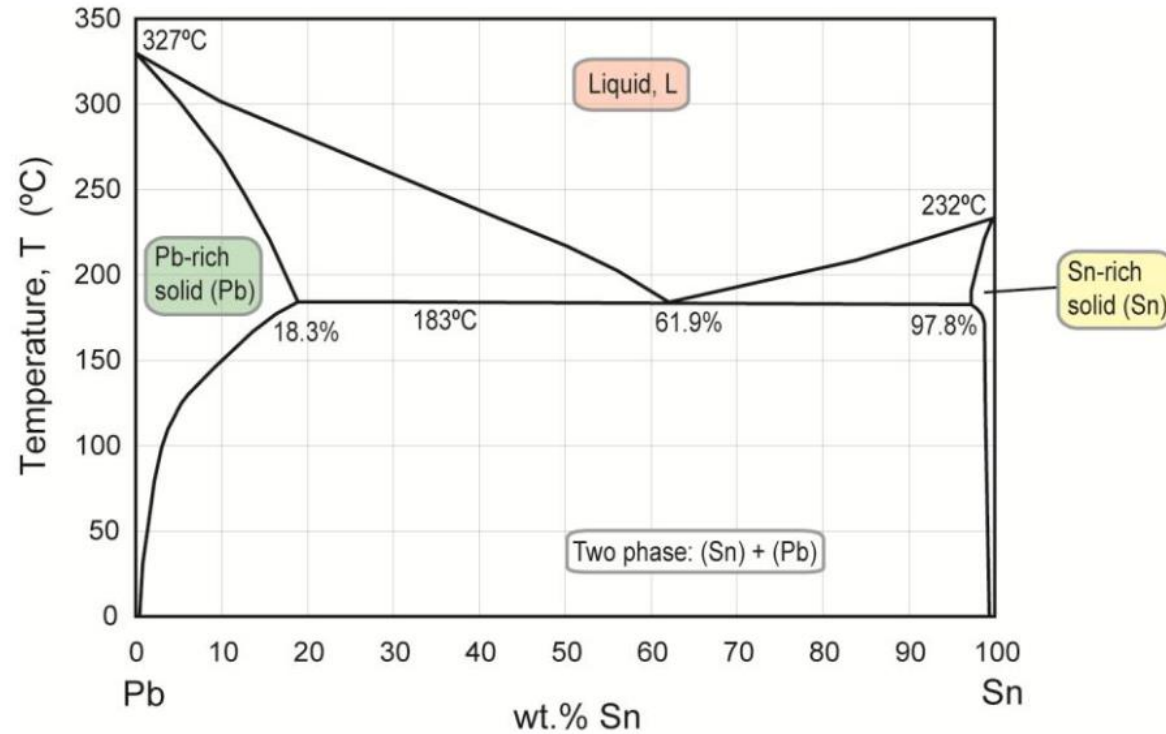
Start – 250°C, α (solid phase rich in Pb),
composition : 5%Sn, 95%Pb

125°C precipitation of β_{α} solid phase (rich in Sn)
due to reduction of the solubility of Sn in Pb at that
temperature

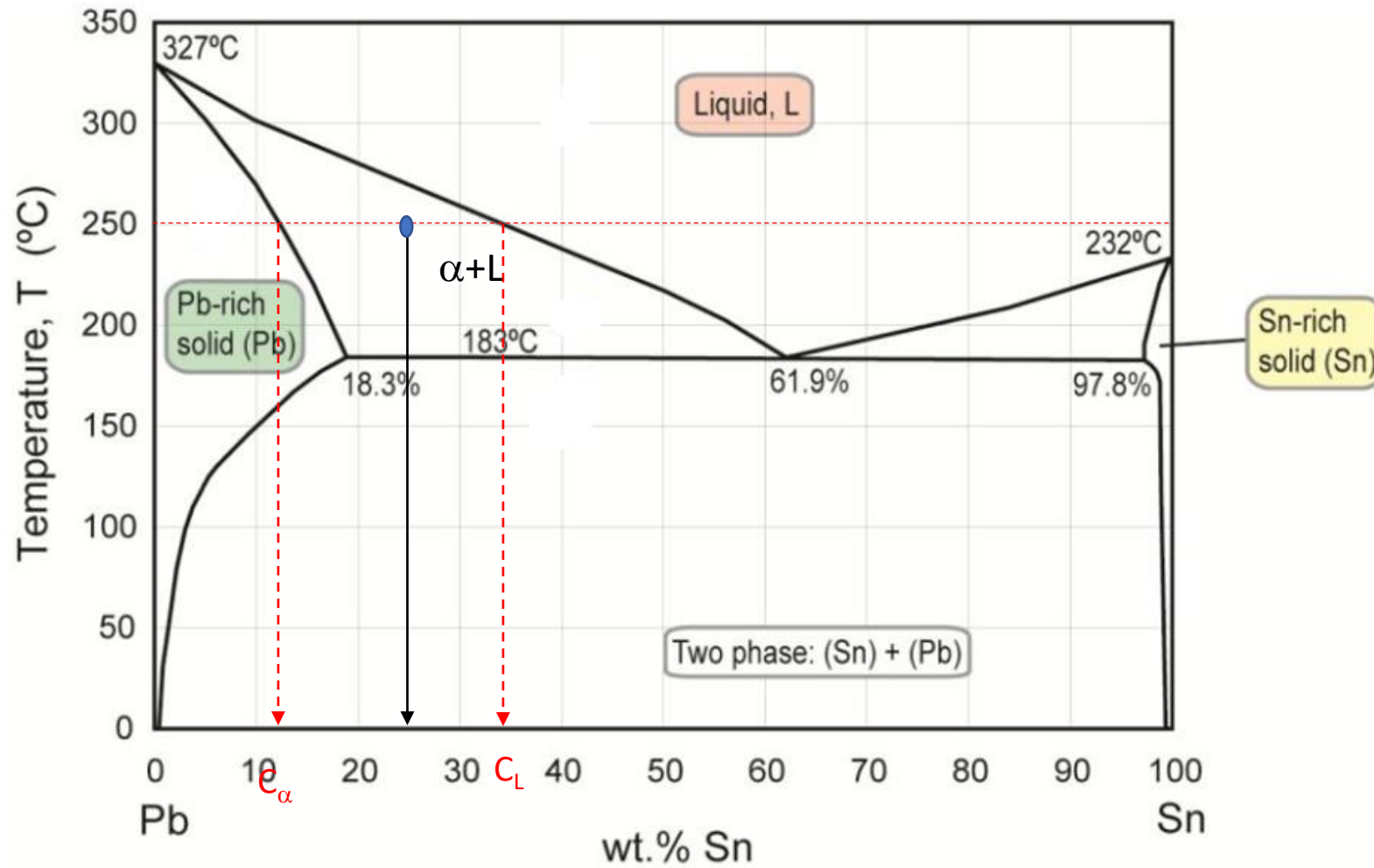
From 125°C to room temperature no other phase
changes occur, the microstructure is solid phase α
(rich in Pb) with small β_{α} crystals



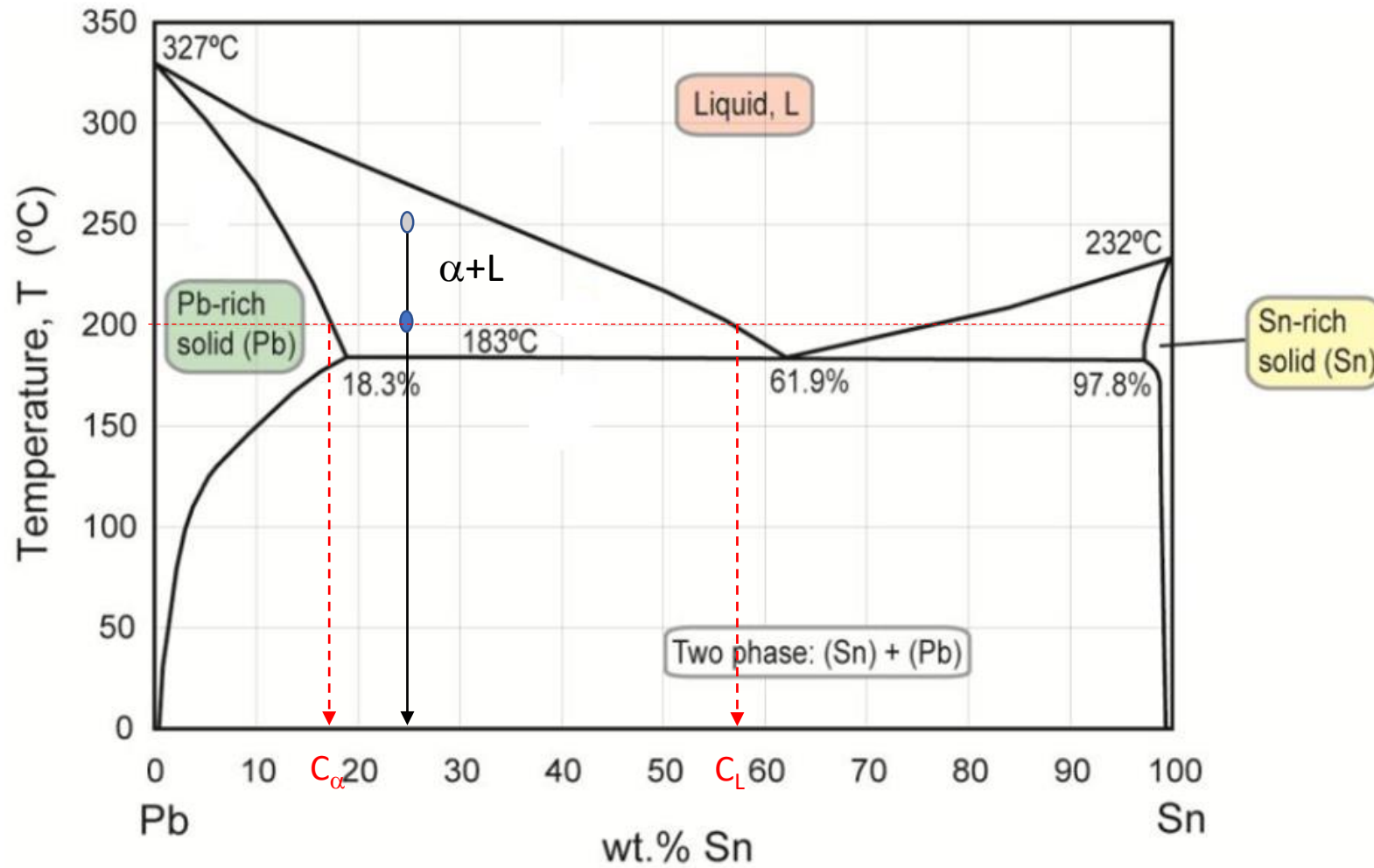
E.7 Use the Pb-Sn diagram below to answer the following questions.



- (a) The constitution point for a Pb-25wt% Sn alloy at 250 °C lies in a two-phase field. Construct a tie-line on the figure and read off the two phases and their compositions.
- (b) The alloy is slowly cooled to 200 °C. At this temperature, identify the phases and their compositions
- (c) The alloy is cooled further to 150 °C. At this temperature, identify the phases and their compositions.
- (d) Indicate with arrows on the figure the lines along which the compositions of phase 1 and phase 2 move during slow cooling from 250°C to 200°C.



Alloy 25%Sn 75%Pb @ 250°C
 α (solid phase rich in Pb) and liquid (L)
Composition of α =12%Sn, 88%Pb
Composition of liquid: 33%Sn, 67%Pb

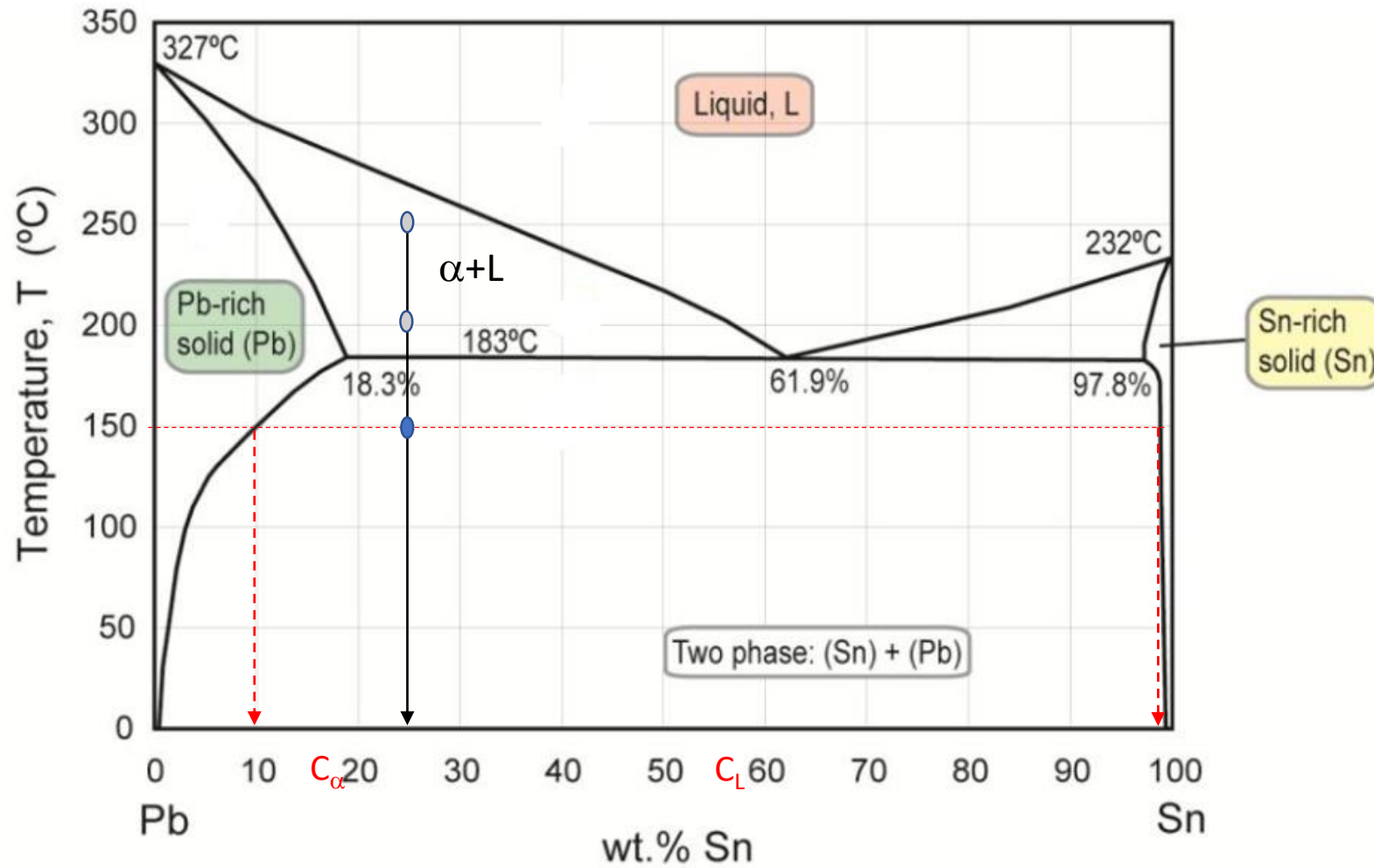


Alloy 25%Sn 75%Pb @ 200 $^{\circ}\text{C}$

α (solid phase rich in Pb) and
liquid (L)

Composition of α =18%Sn,
82%Pb

Composition of liquid: 58%Sn,
42%Pb

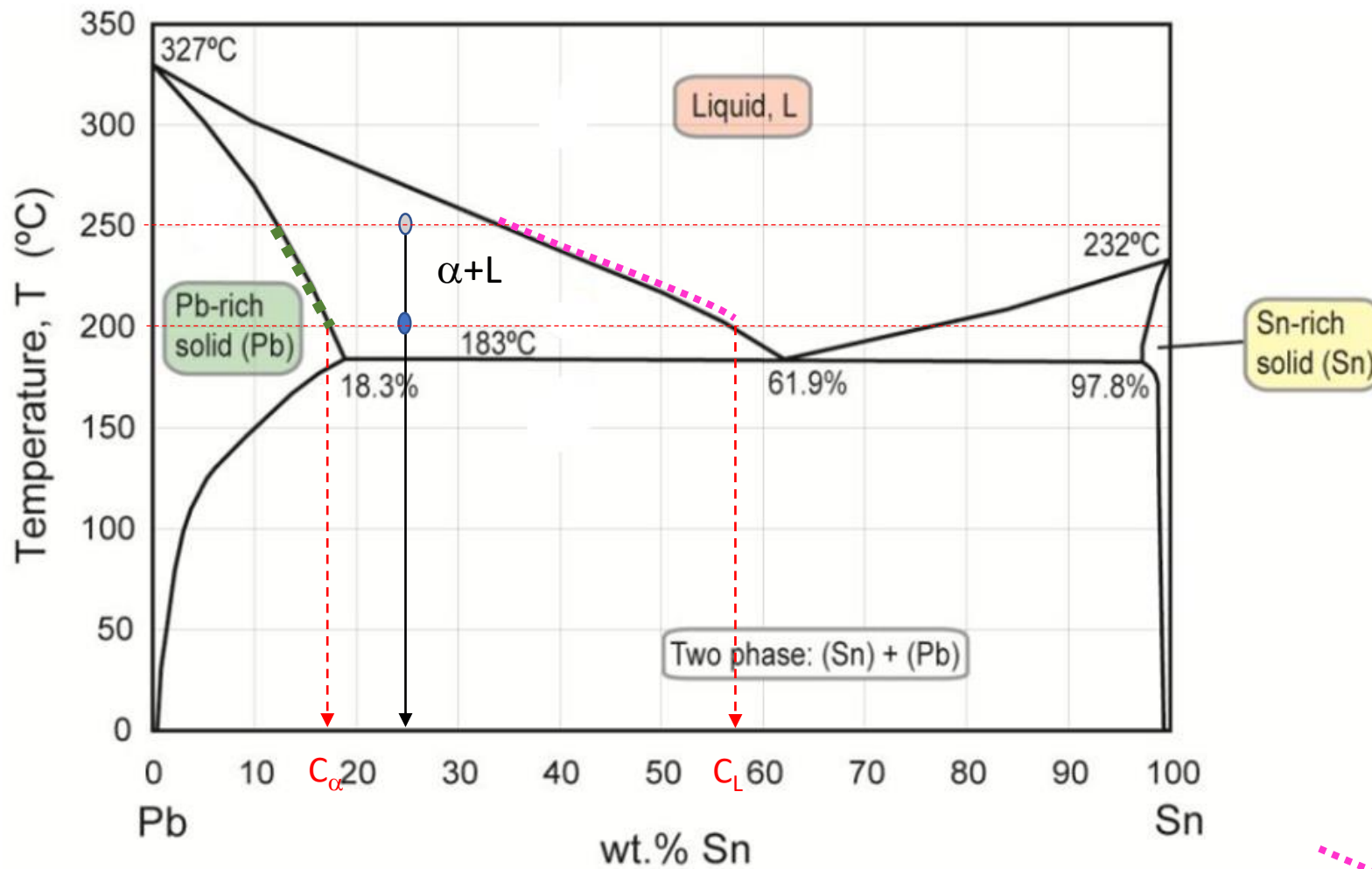


Alloy 25%Sn 75%Pb @ 150 $^{\circ}\text{C}$

α (solid phase rich in Pb) and
 β (solid rich in Sn)

Composition of α : 10%Sn,
 90%Pb

Composition of β : 99%Sn,
 1%Pb



Alloy 25%Sn 75%Pb @ 250 $^{\circ}\text{C}$

α (solid phase rich in Pb) and liquid (L)

Composition of α =12%Sn, 88%Pb

Composition of liquid: 33%Sn, 67%Pb

Alloy 25%Sn 75%Pb @ 200 $^{\circ}\text{C}$

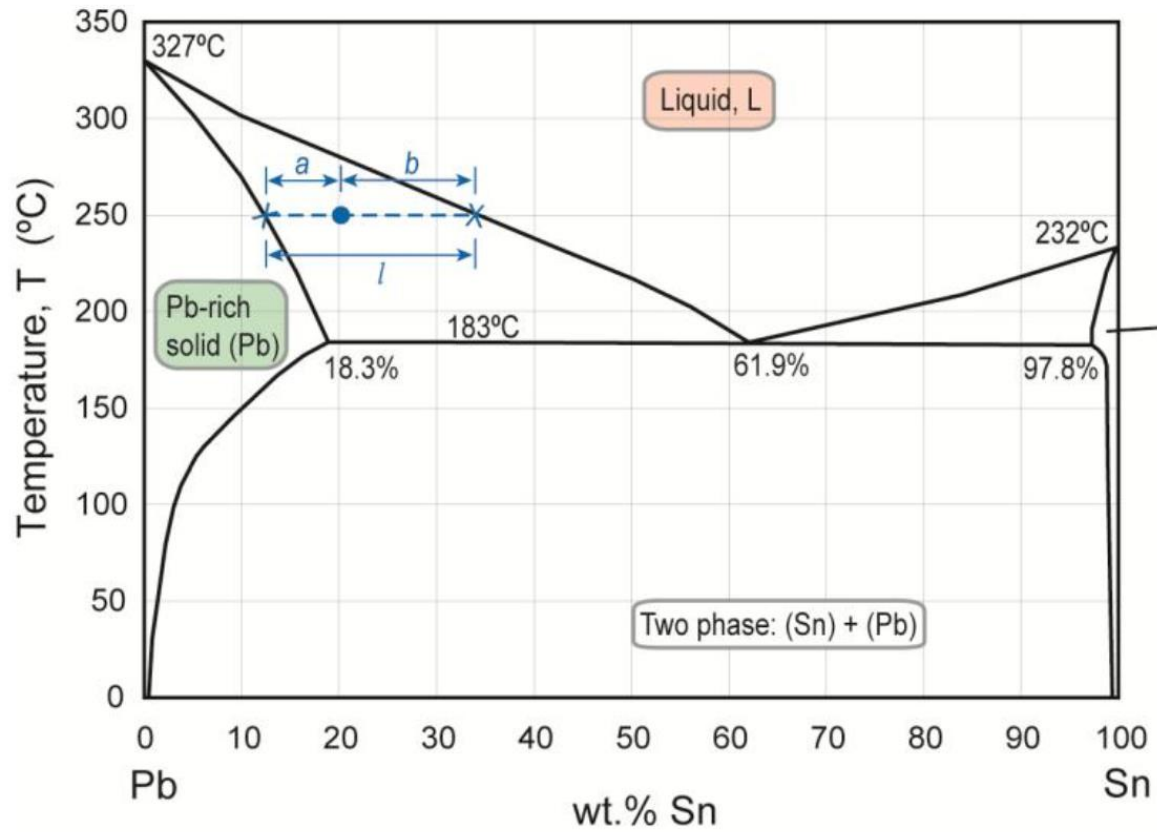
α (solid phase rich in Pb) and liquid (L)

Composition of α =18%Sn, 82%Pb

Composition of liquid: 58%Sn, 42%Pb

Variation of α composition reducing temperature from 250 to 200 $^{\circ}\text{C}$

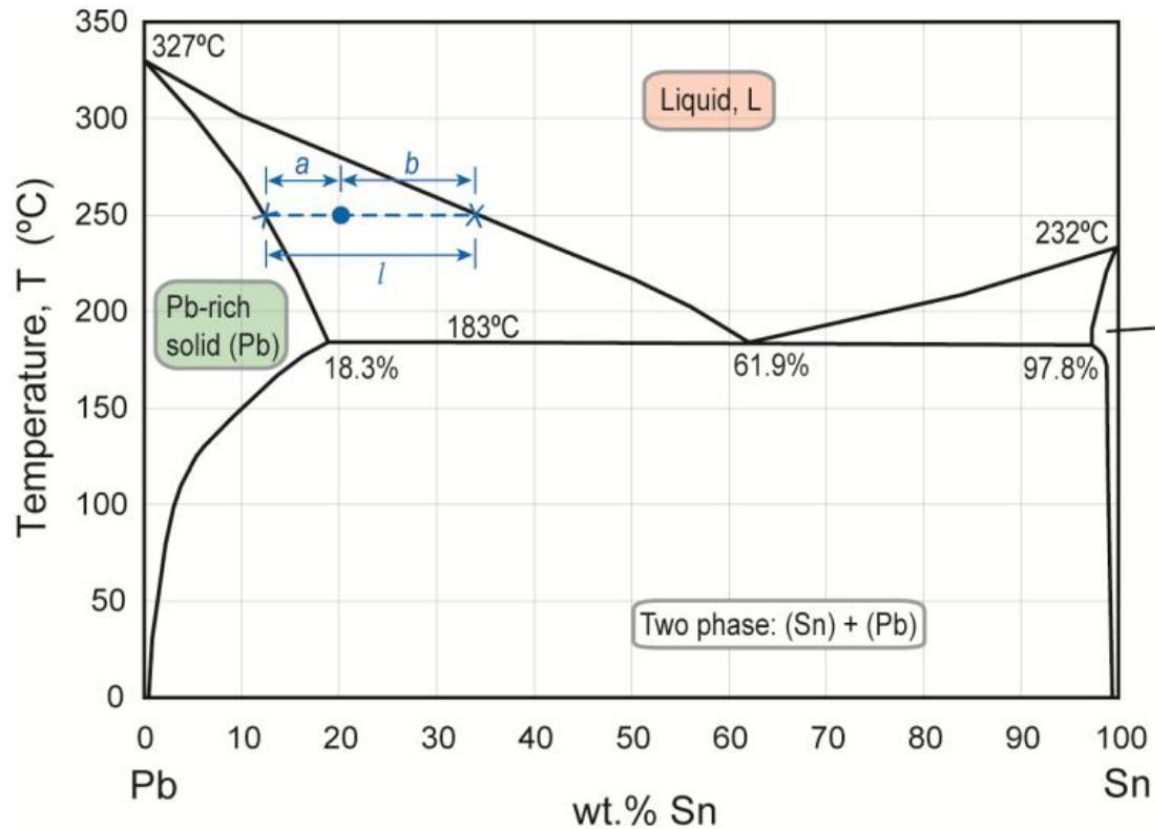
Variation of L composition reducing temperature from 250 to 200 $^{\circ}\text{C}$



Consider Pb-20wt% Sn at 250 °C in Figure

A constitution point in the two-phase field: liquid plus Pb-rich solid.

Find the proportions of each phase



Consider Pb-20wt% Sn at 250 °C in Figure

A constitution point in the two-phase field: liquid plus Pb-rich solid.

To find the proportions of each phase

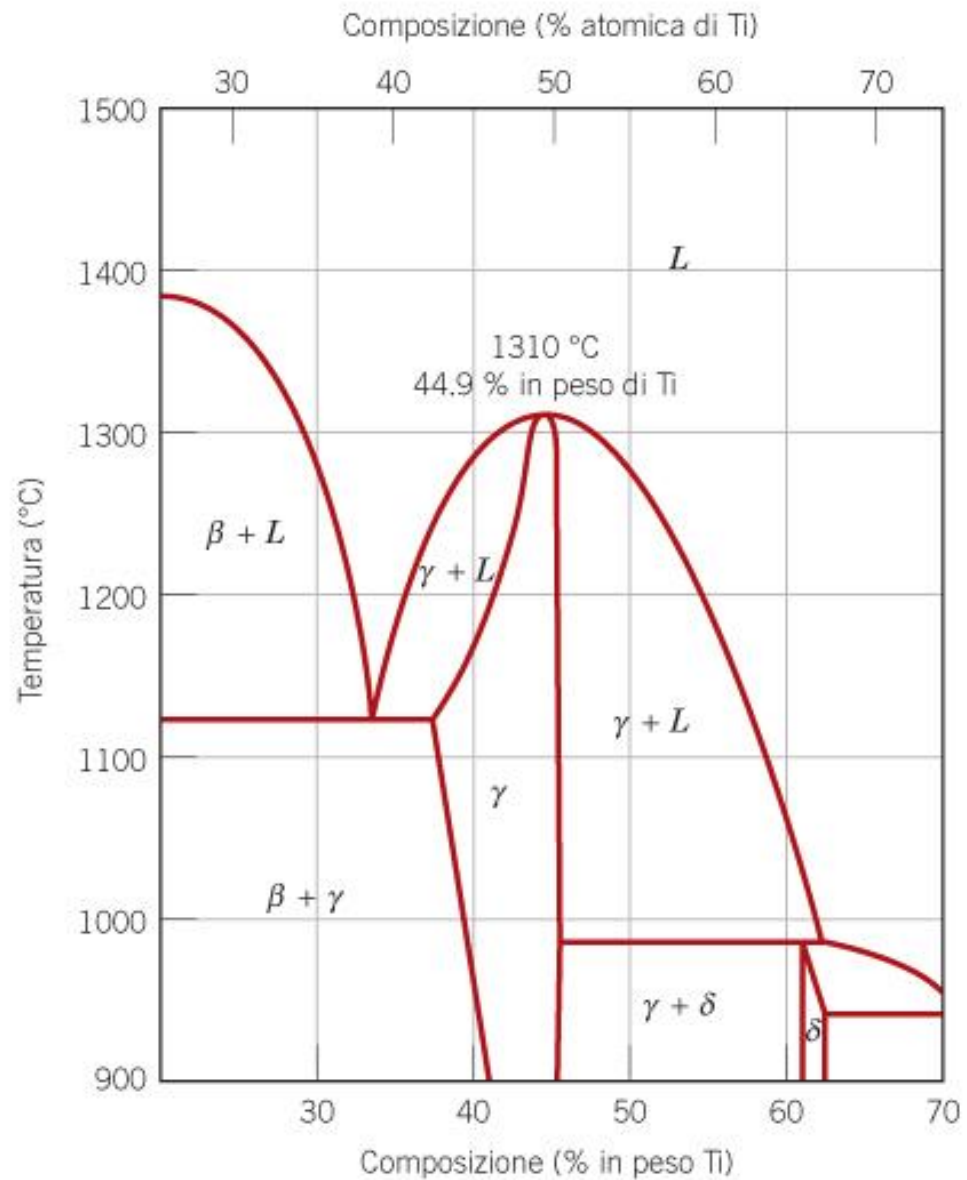
First construct a tie-line through the constitution point and read off the compositions of the phases:

Pb-rich solid with $W_{Sn}^{SOL} = 12\%$; *Liquid* with $W_{Sn}^{LIQ} = 34\%$.

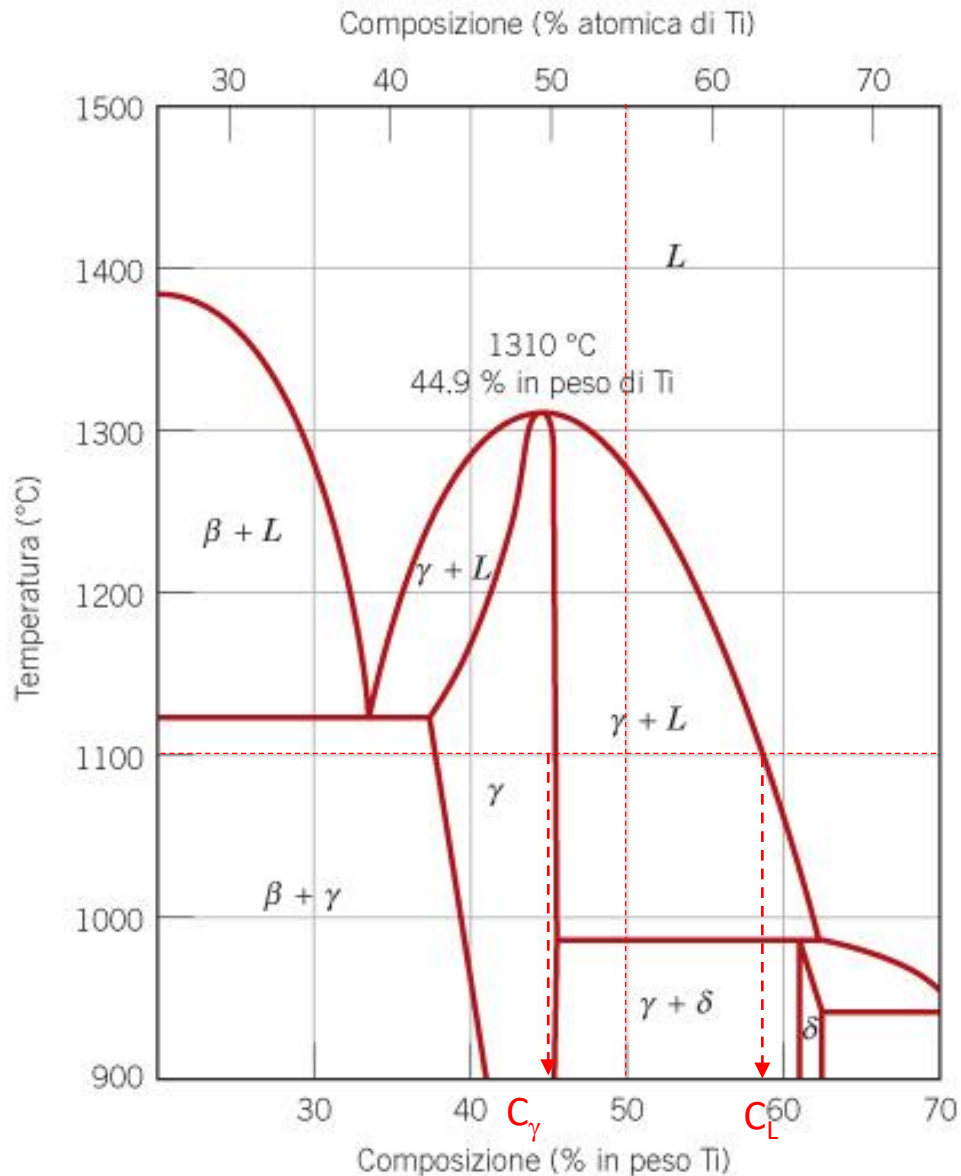
The tie-line is of length l , while the lengths of the segments to either side of the constitution point are a and b respectively

For the example alloy of composition $W_{Sn} = 20\%$

Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.



Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.



Phases: solid phase γ (intermediate solid solution) and liquid L

Compositions:

Composition of γ : 45%Ti

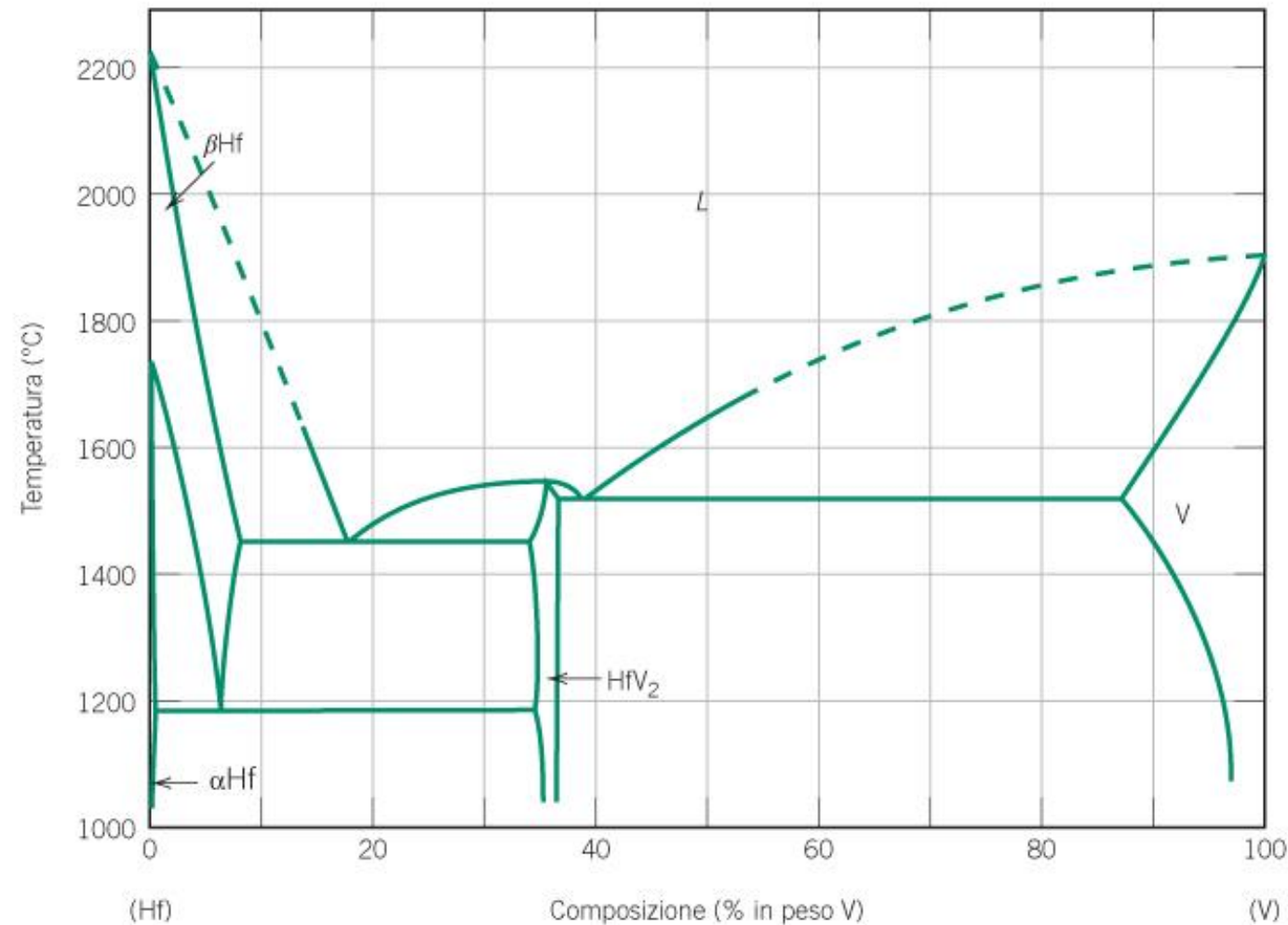
Composition of Liquid: 58%Ti

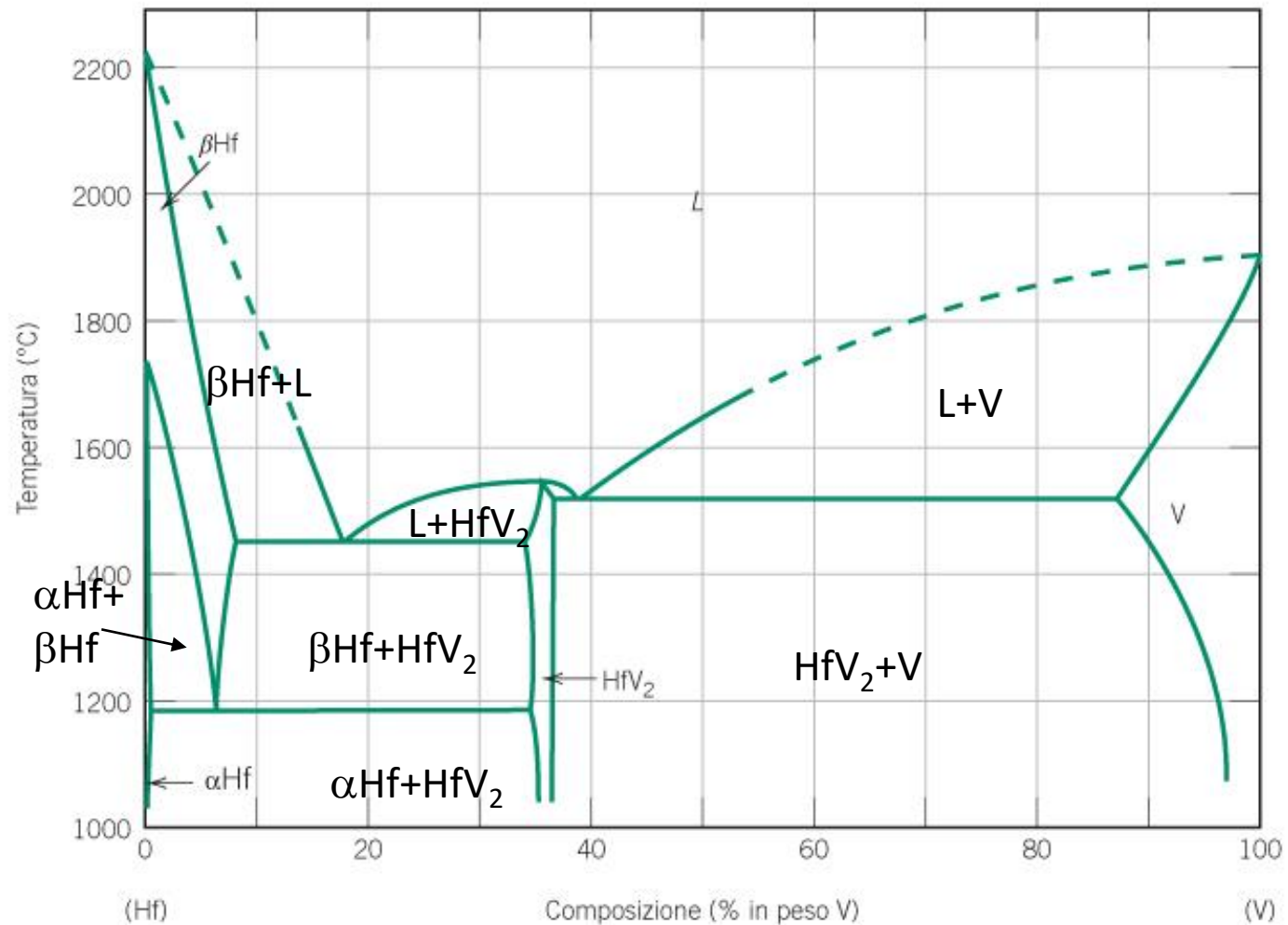
Relative amounts

$$\% \gamma = (58 - 50) / (58 - 45) * 100 = 61.5\%$$

$$\% L = (50 - 45) / (58 - 45) * 100 = 38.5\%$$

Complete the following diagram





Complete the following diagram

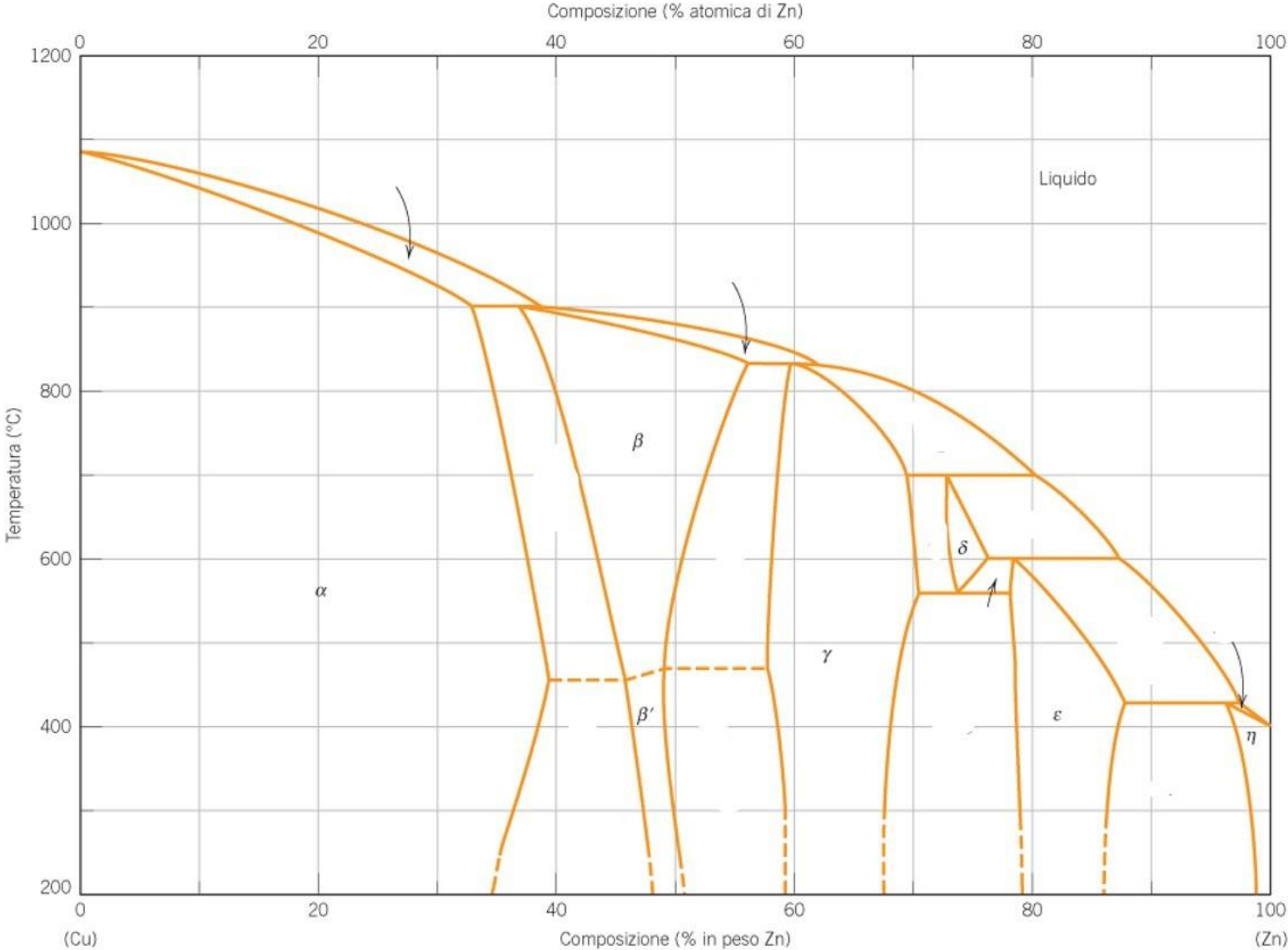


Figura 9.19 Diagramma di fase rame–zinco.



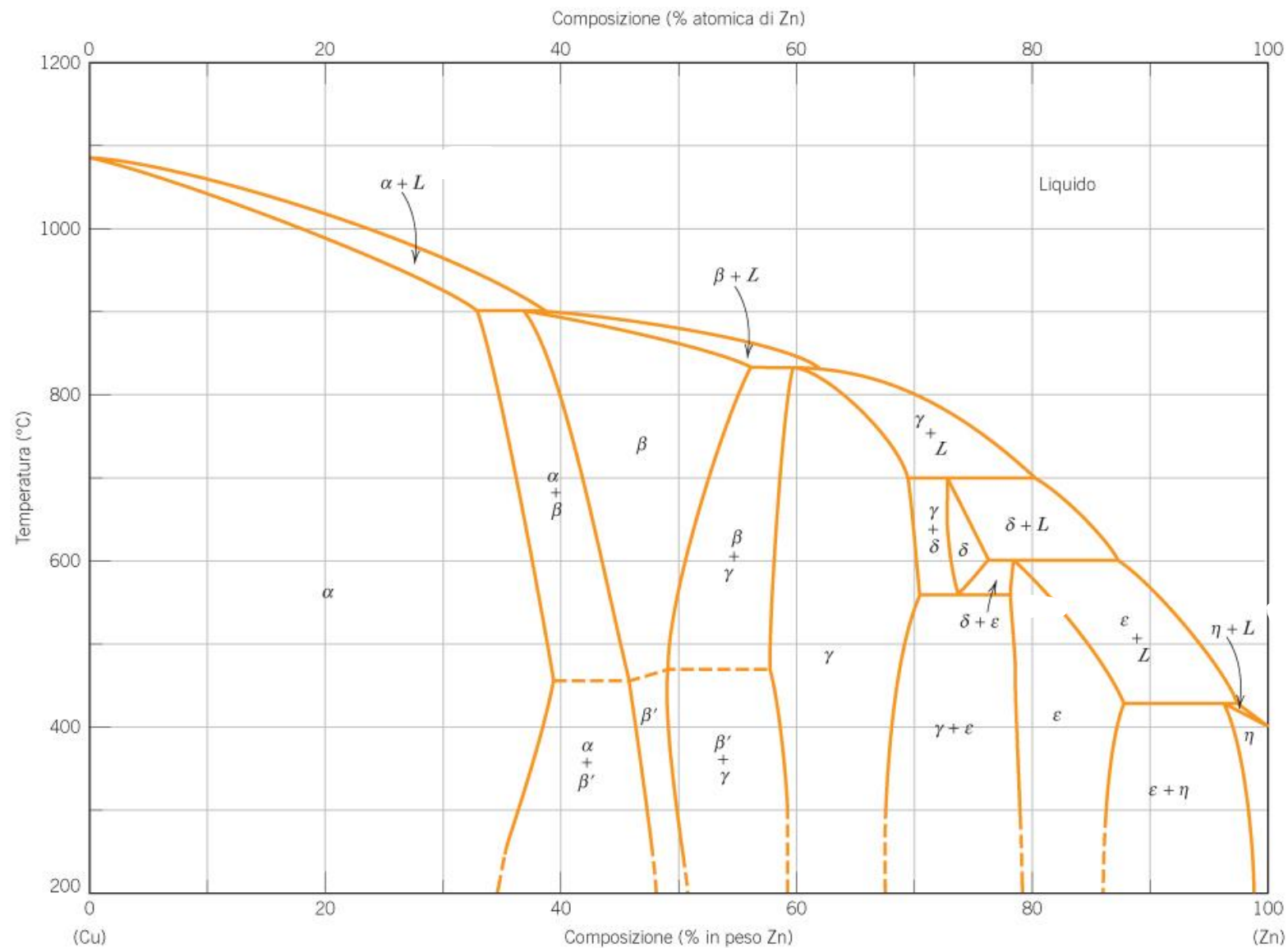


Figura 9.19 Diagramma di fase rame-zinco.